FORERMED

PZDR User Manual

V 2.0.14

Revision History

Version	Modifier	Content	Modification time
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V2.0.14		Modify	2023/11/10

Contents

1. Introduction

1.1 Summary

This document is the manual of PZDR software from FORERMED Medical Co, Ltd.

This document is intended for DR equipment manufacturers, DR film technologists, and DR diagnosticians. The article describes in detail the functions of the PZDR software, including the interface design and workflow of the software, and clearly explains how to use PZDR for routine filming and diagnostic work, as well as the detailed operation steps.

Before using the software, please read the manual carefully. PZ Medical shall not assume any legal liability for any damage caused to people or equipment due to its failure to perform operations in accordance with the requirements of this manual.

1.2 System Use Scope and Contraindications

PZDR needs to be used with the flat panel detector of FORERMED Medical and can be connected with all static flat panel detectors. PZDR needs to be connected to the flat panel detector and set the type of connected flat panel detector when using.

No contraindications.

1.3 System Function Overview

This PZDR software is medical software used in conjunction with our detector and is a comprehensive filming and diagnostic system developed to facilitate doctors' diagnoses and patients' consultations. Using PZDR, doctors can complete the acquisition, transmission, and film printing of X-ray examination pictures of patients and other examination items. The multiple picture processing methods embedded in the software also greatly facilitate doctors' consultation and can significantly improve the speed and accuracy of consultation.

PZDR includes the following functional modules：

A. Examination registration management

Patient information is registered and stored in the database, which is associated with the images of patient examinations.

B. Film examination

According to the registered examination site, the film examination is carried out, and the body position can be added or deleted at any time to adjust the film dose.

C. Dual detector switching

Two flat panel detectors can be connected to switch freely for exposure processing.

D. Diagnostic aid

Enhance the DR image, and provide a series of tools to assist diagnosis such as transferring, cropping, and marking.

E Case management

Patient cases and examination images are stored in the database and can be browsed at any time.

F. System integration

Adopting standard DICOM 3.0 communication protocol, it can easily connect to hospital HIS, RIS, PACS, and other systems, and connect to a DICOM film printer.

G. Film Printing

Support local printing and DICOM network printing.

H. Diagnostic Report

Reports can be set up according to the needs of hospitals or departments, and the report layout can be edited.

2. Software Installation

2.1 Installation Environment

The computer requirements of this workstation software are shown in the table below.

	Minimum configuration	Recommended configuration
CPU	Intele Eighth generation i5 / i7 or higher	
Memory	8G or more	And 16GB or above is recommended
Hard Disk	512G or above	Greater if recommended without an image storage server
Network	TCP/IP	At the same time, to meet the network requirements of the detector
USB mouth	At least one, for hdog	Except for pure software dogs
Ethernet card	Pro 1000M	Pro 1000M
Display resolution	Resolution 1920x1080 color display	Adaptive 2K or 4K for larger resolution
Operating System	Windows 7/10	Windows 10/11

2.2 Installation Procedure

Versions prior to V2.0.14_20220810 (includes V2.0.14_20220810)：

- Software installation package, double-click the install
- Select the installation language
- Follow the prompts to install dot Net Framework 4.6 and vc_2015, skip if already installed
- Select the installation path, generate a desktop shortcut, confirm and start the installation and display the installation progress
- Confirm completion

Versions later than V2.0.14_20220810：

- Select the installation language
- Select the installation path, generate a desktop shortcut, confirm and start the installation and display the installation progress
- Confirm completion
- Open the desktop "PZDR" folder, open the folder "runtime", and choose "vc2015_x86" for installation, has been installed, skip.

3. Introduction of Software Usage

3.1 Software Configuration

A. Configure the current detector type

The software automatically recognizes the type of flat panel detector without setting. Set this parameter when switching between two detectors, see the description in 4.12 for details.

B. Configure detector working mode

HST is the synchronization mode, AED is the exposure auto-detection mode, and TEST is the simulation on the map mode. HST and AED mode work need to refer to the flat panel detector data sheet and user manual, TEST mode is the simulation on the map mode, need to connect the detector, the default upload chest film orthostatic images, do not need X-rays.

C. Generator linkage mode

The default TEST mode is not linked with the generator, if you need to link, select the corresponding high voltage name in the drop-down bar.

D. Other settings

Other settings of PZDR are hospital information, department information, DICOM node information, film information, etc. Refer to the 4.6 Configuration tool description for details.

3.2 Registration

1. For General examination, enter patient-related information, add body position, and click the examination button to enter the examination interface.

2. Registration examination, enter patient-related information, add body position, click the registration button to register the patient's examination information to the task list and select the patient examination that needs to be exposed in the task list.

3. Emergency, the default patient information is provided by the system, and the

operator selects the body position and starts the examination.

3.3 Registered Inspection

1. Go to the task list and quickly search for the corresponding inspection by name, number or time.

2. Check, select the inspection that will be performed, click the check button to start the inspection.

3. Delete inspection, check the inspection that needs to be deleted, click the delete button to delete the inspection, and at the same time, there is an option to delete the registered inspections in bulk.

3.4 Transmission Queue (PACS, WorkList)

1. send queue, showing the current status of all exams being sent from DR to PACS.

2. task list, containing patient information from RIS or DICOM stations.

3.5 Completed Inspection

1. Enter the database list, and quickly search for the corresponding test by name, test number or time.

2. Modify the information, select a test, click the Modify button to modify the patient's name and other information, and then click Refresh to update the patient information.

3. Read the image, select the inspection to be done, and click the Browse button to

start reading the image.

4. Delete inspection, check the completed inspection and click the Delete button to delete the completed inspection in batch.

5. Send images, check the completed inspection, and click the Send button, you can batch send the images of the completed inspection, the sent images will be in the PZDR send queue, and the task will send the images to the image server one by one.

6. Print image, select the inspection to be printed, and click the Print button, you can switch to the film printing interface, the image will be laid out for printing.

7. Review, select an examination, click the Review button, and enter the Prepare Exposure interface to re-expose and reprocess the body position.

8. Report, select an examination and click the Report button to enter the Edit Report interface.

9. Burn, check the examination to be burned, and click the burn button to burn the examination to CD.

10. Export, check the examination to be exported, click the export button, select the path to be exported in the dialog box, and click the confirm button to complete the export operation.

3.6 Exposure Interface

1. Enter the examination interface, and add and delete examination positions.

2. Dose, each body type, each position of the pose to use a different dose of

radiation to shoot, the software database saves a set of default dose data, when the body type is selected, position placement, the dose control area will display the corresponding generator parameters, if needed to adjust, it can be adjusted through the plus or minus button, it can also be saved through the save button, the adjusted reference dose saved to the database.

3. Quickly modify the dose, the reference dose can be quickly modified by changing the body type.

4. Device status, the inspection interface will indicate the plate status, and the generator status, the doctor can control the exposure according to the status indication.

5. After exposure, the detector will transmit the image to the PZDR software, the software will do the basic processing of the image, enhance the image for the set shooting position, and then display the processed image in the acquisition interface.

6. The acquisition interface provides a series of adjustment tools to assist diagnosis, such as shift, zoom, mirror, rotate, brightness contrast adjustment, area of interest adjustment, marker, crop, etc.

7. If dissatisfied with the acquired image, the image can be rejected and re-taken without deleting the body image.

8. After clicking Save, this shooting is completed and the software will store the image and generate a thumbnail image to replace the previous pose image.

3.7 Image Viewing and Reprocessing

1. After finishing the inspection, review the image, the image can be re-processed,

and there are also auxiliary diagnostic tools for adjusting the image, moving, scaling, mirroring, rotating, brightness and contrast adjustment, area of interest adjustment, marking, cropping, etc.

2. When reprocessing, there are four processing effects for reference, and certainly free to adjust the post-processing parameters to adjust the image to a satisfactory effect.

3.8 Print Layout

1. In the "Database" interface, multiple cases can be selected for printing, and images from multiple cases can be printed at once by including them in the list to be printed.

2. Set the film size to be used for printing in the configuration file, and the print module will import the adjusted images into the film area.

3. The print interface provides a series of tools to adjust the printed image, including shift, zoom, mirror, rotate, brightness and contrast adjustment, area of interest adjustment, etc.

4. After clicking Print, the software sends the film to a printer, either a local printer or a DICOM standard printer.

3.9 Diagnostic Report

1. The doctor gives the description and diagnosis of the disease according to the image, while the description and diagnosis of the disease in the system can assist the doctor to complete the diagnosis, note that the diagnosis is for reference purposes only.

2. After completing the diagnosis, the report can be generated, in addition to the description of the disease and diagnosis, the report can also add images, preview the report, and print the report after the doctor confirms that it is correct.

3. Doctors can use the report template editing tool to make custom changes to the initial report template.

4. Software Function Description

4.1Login Interface

Figure 4.1-1 Login interface

Login interface, the default user is Admin, the password is 1, configuration interface can modify, add and delete users.

4.2 Initialization Interface

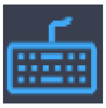
Button	Description
	Login to the PZDR system
	Cancel login
	on-screen keyboard

Figure 4.2-1 System Initialization

Before entering the registration interface, the system needs some preparation work, first initialize the detector, detect the connection status of the detector, serial number, etc., and then initialize the database, import all the information in the database into the interface, and then initialize the generator. Preparing all this the system starts normally. Check the current status of the system according to the progress bar.

Hold down the CTRL + ALT combination key to move the detector initialization progress bar coordinate operation. After moving to the determined position, press the CTRL + S combination key to save.

（Note: The detector initialization progress bar moves after pressing the CTRL + ALT key, and the progress bar will not continue to load the data. Press the CTRL + S key to save the progress bar to load the data.）

4.3 Registration Interface

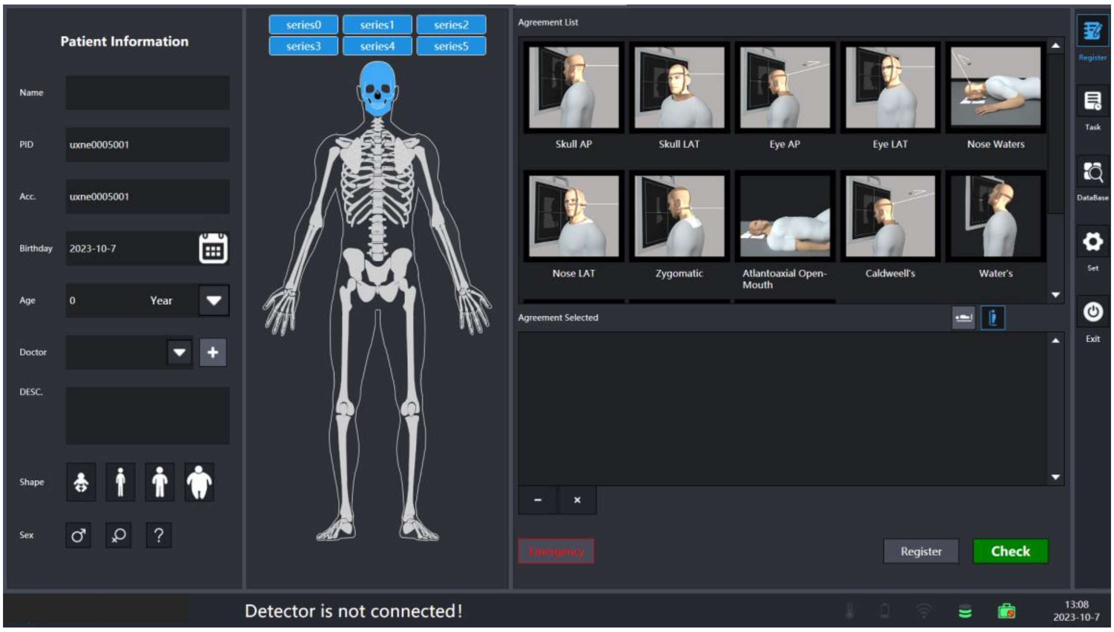


Figure 4.3-1 Registration Interface

There are five symbols on the left corresponding to these buttons: Register, Task, Database, Set and Exit.

Button	Function

	Register the button, click to enter the registration page
	Task button, click to enter the task list
	Database button, click to enter the database list
	Set the button, the dialog box appears, enter the password (the password is the login password:1), enter the configuration tool interface
	Exit button, exit the software

The bottom row is the software logo, detector temperature, detector power, the strength of the signal connected to the detector, hard disk information, and detector serial number. As shown in the following table:

Symbol	Description
FORERMED	Software logo. Click this symbol to check the current software version.
	Battery Indicator. click to view the remaining detector power.

	Signal state. Check the intensity variation of the signal from 0-4.
	Detector state. Click to check the current detector serial number and temperature.
	Hard drive remaining space. Check hard drive space remaining.
	Check the current temperature of the detector.

Add the patient's name, patient number, examination number, birthday, age, doctor, examination description, body type, gender and other information, where the patient's name, patient number, and examination number are required information. After filling in the information, select the part on the body diagram with the mouse and left click on the part, all the optional positions corresponding to the part will be listed in the protocol to be selected, click on the position in the protocol to be selected, the software will automatically add the position to the selected protocol. The first button below the selected protocol can delete the selected body position, and the second button clears all the body positions. As the following table:

Button	**Function**

−	Deletion position
×	Empty position

In emergency, the system provides default patient information, the operator chooses the position, quickly enters the exposure interface, and carries out exposure; After adding the positions click Register to add the patient information to the task list, or click Check for exposure. The following table:

Button	Function
Emergency	Urgent Care button to quickly add patient information.
Register	Enter patient information into the task list.
Check	Access the exposure interface.

4.4 Task List

4.4.1 Task List

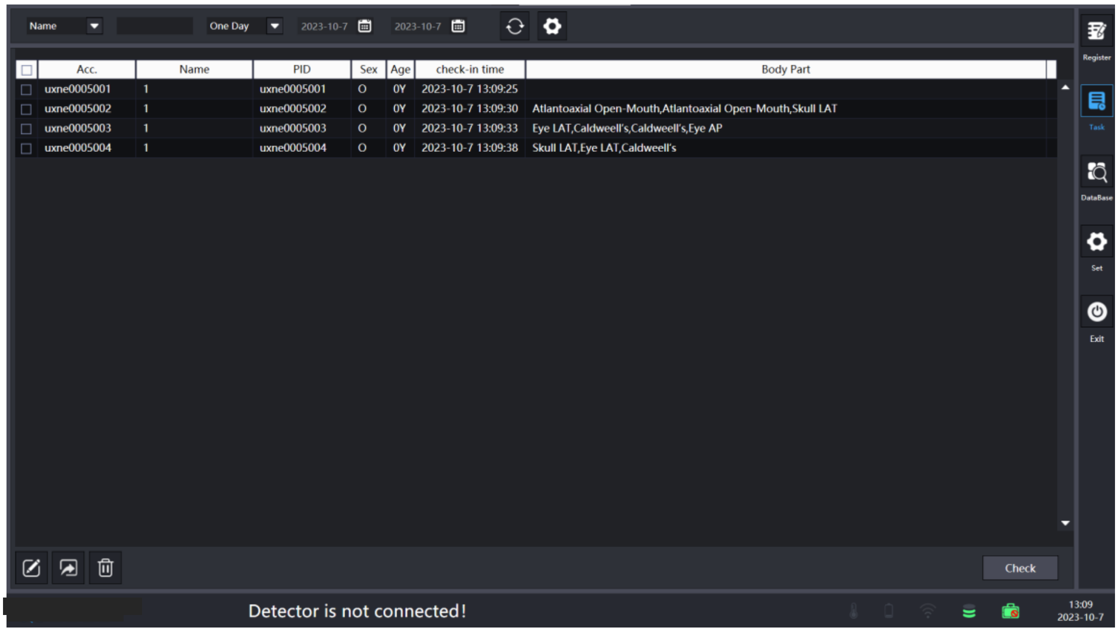
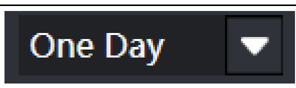
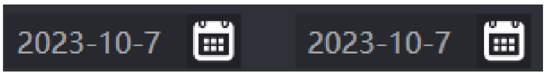
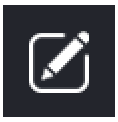
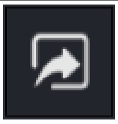
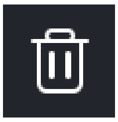


Figure 4.4-1 Task List

The information after each registration will be stored in this list. Click the Refresh button above to filter the information; the Set button to select the case display information; the Modify button to modify the name, gender and other information of this examination; the Delete button to delete the selected information; the Job List, where the examinations registered by other systems (RIS) are directly imported into the task list of the software via network transmission.

Button	**Function**
 	Set the search criteria, fill in the blank field for keyword search.
	Set search periods.

	
	Refresh button to search for all patients who meet the criteria.
	Settings button, set the information of the case displayed below.
	Modify button to modify patient name and other information.
	Import button, select ImportDemo.xlsx under the software installation path to import cases to the task list in bulk.
	Delete button, click to delete the selected patient information.

When exposing, click the check button to enter the exposure interface.

Button	**Function**
	Check button, select a message and click to enter the exposure interface.

4.5 Database

4.5.1 database

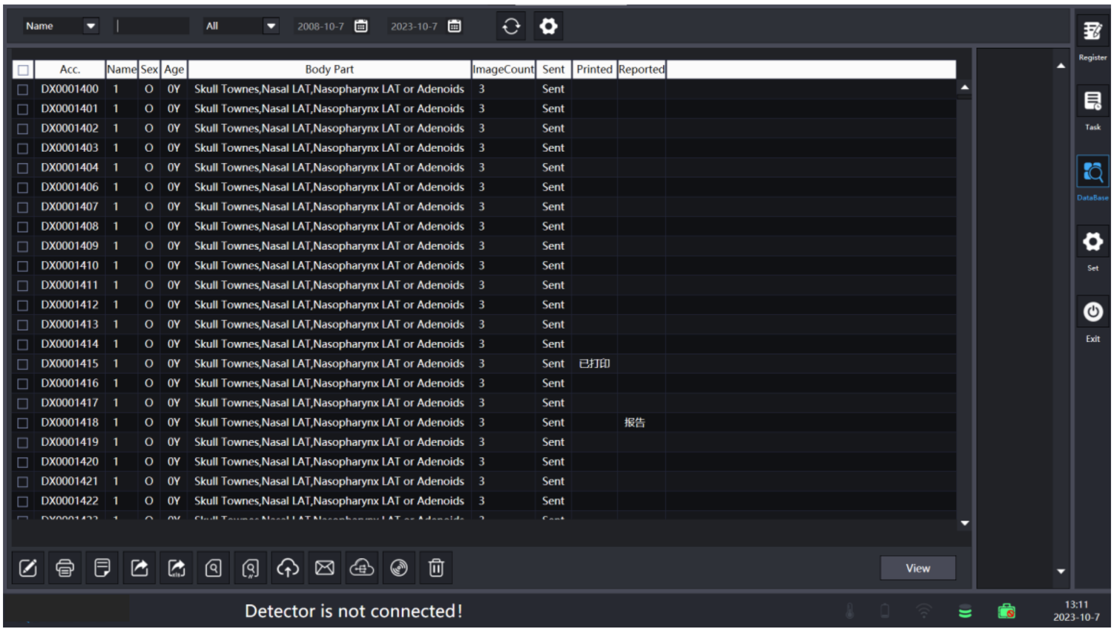
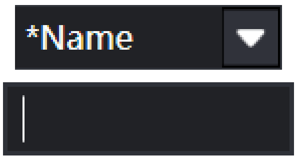
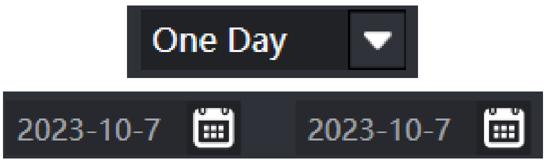
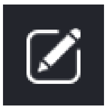
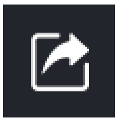
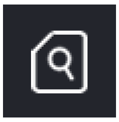


Figure 4.5-1 Database

The completed inspections are stored here. The upper refresh button filters the exams according to the conditions; the settings button displays the case information below according to the selected fields. The lower buttons allow you to modify, send to PACS, print, report, review, export and delete the selected exams respectively, and the view button at the bottom right allows viewing the images of the exam when clicked.

Button	**Function**
	Set the search criteria, fill in the blank field to search for keywords.
	Set the search time period.
	Refresh button to search for all patients

	who meet the criteria.
	Set button, select the information field of the case to be displayed below.
	Modify button, click to enter the information modification interface.
	Print information, click to enter the print interface.
	Report, click to enter the report interface.
	Export, export the selected examination to the specified directory.
	Excel export, export the selected examination to the specified interface in the form of Excel.
	Review, click to reprocess images.
	Review, create a new examination with the same information as the selected patient.
	Send the information to the serve.

	PACS Queue, check the status of PACS sending.
	Burn, burn the selected examination to CD.
	Delete button, click it to delete selected information.

View information, click View to enter the viewing interface.

Button	Function
	View button, click to enter the browse interface.

4.5.2 Information modification interface

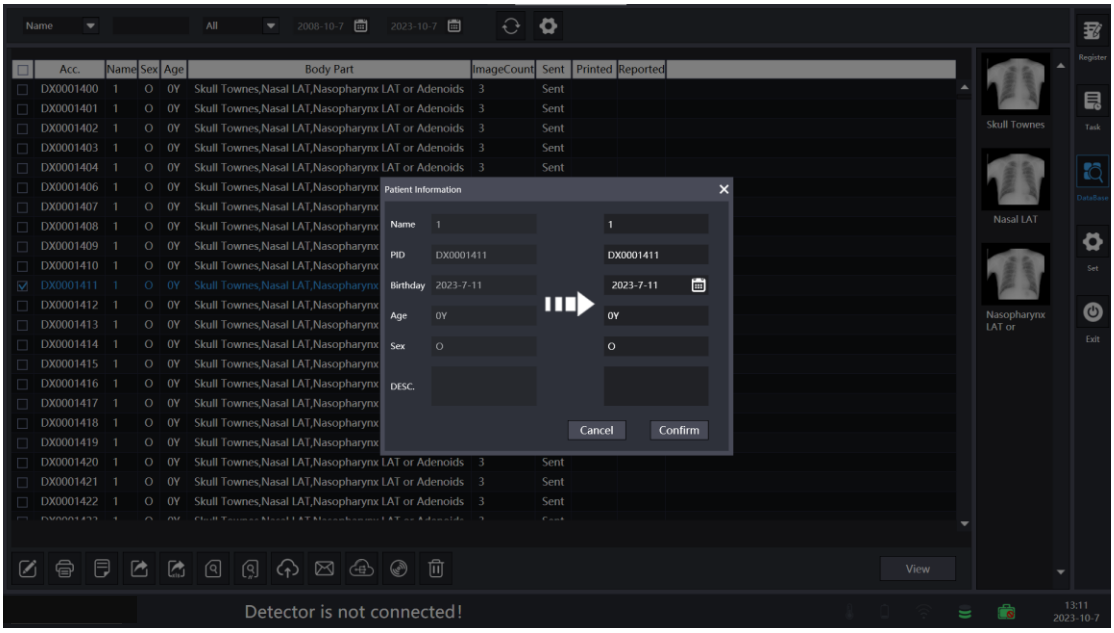


Figure 4.5-2 Modify Information

As shown in the figure, this interface is used to modify name, number, birthday, age, gender, and description; the left side is the content before modification, the right side is the content after modification, and the two buttons are back and save modification respectively.

Button	Function
Cancel	Return button, click to cancel the modification.
Confirm	Save button, click to save the modified information.

4.5.3 Burn Interface

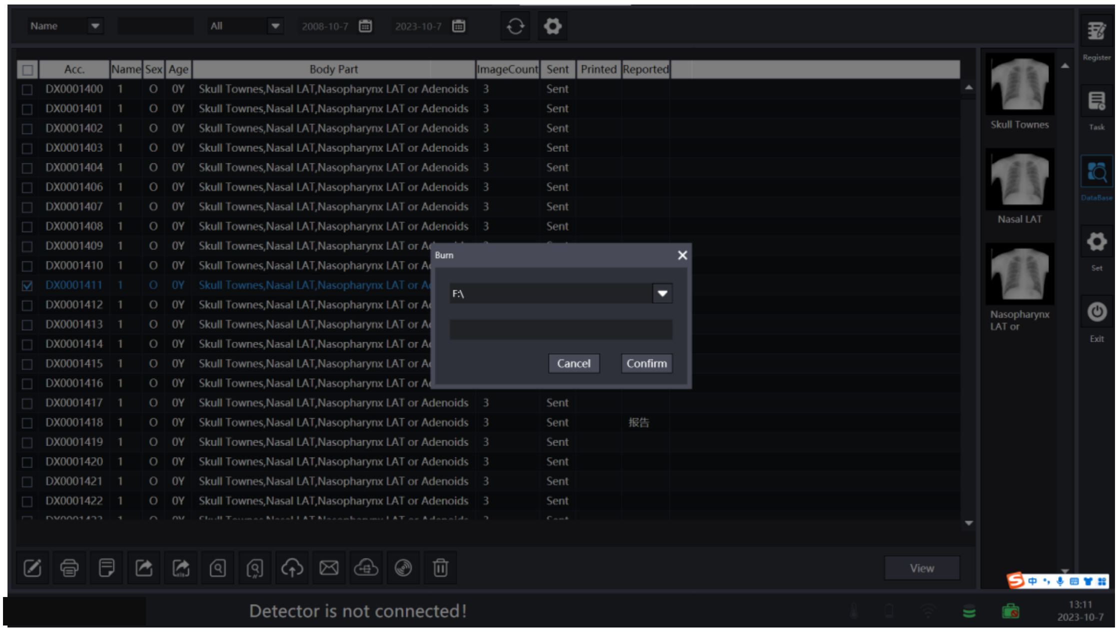


Figure 4.5-3 Burn

Select the burner and the status bar at the bottom will show the current burning progress while burning, currently there are two states, burning and completed burning.

Button	Function
Cancel	Return button, click to return to the previous screen.
Confirm	Save button, click to start burning, already burned click invalid.

4.5.4 Export Interface

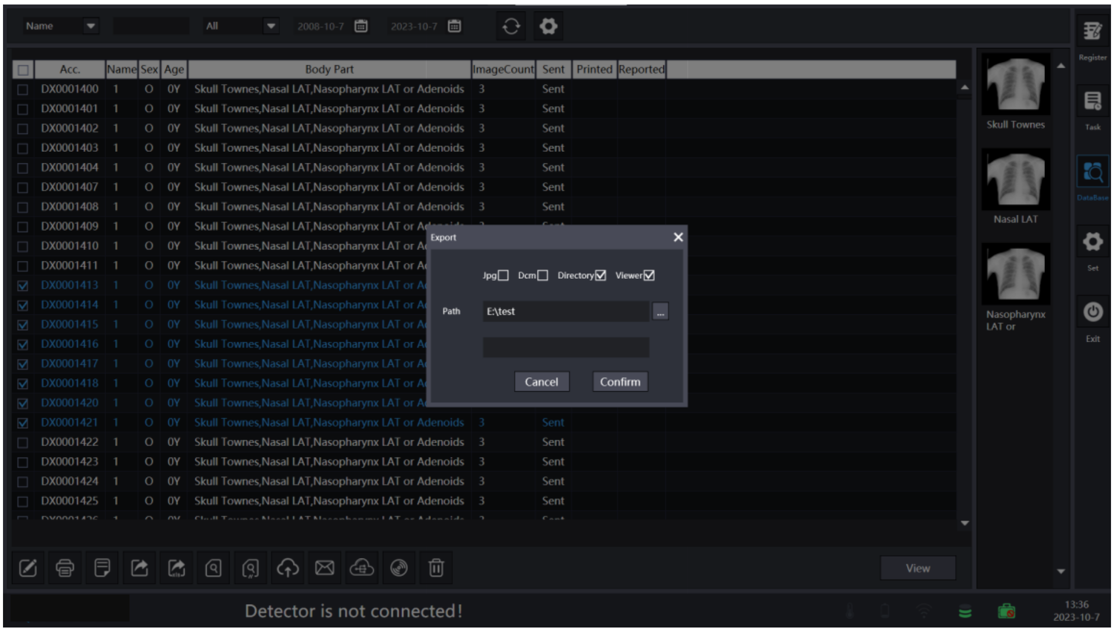


Figure 4.5-4 Export

Select one or more pieces of information in the database list, click the Export button, the above dialog box pops up, select the export path, click the check mark button to complete the export operation, and the dialog box below the path shows the current export status.

Button	**Function**
Jpg ☑	Only images will be exported in JPG format when checked
☑ Dcm	Only images will be exported in DCM format when checked
☑ Viewer	Export the complete database image file after checked
...	Click the button to select the export path
Cancel	Return button, click it to return to the previous interface
Confirm	Save button, click it to start burning, already burned click invalid.

4.5.5 Excel leading-out

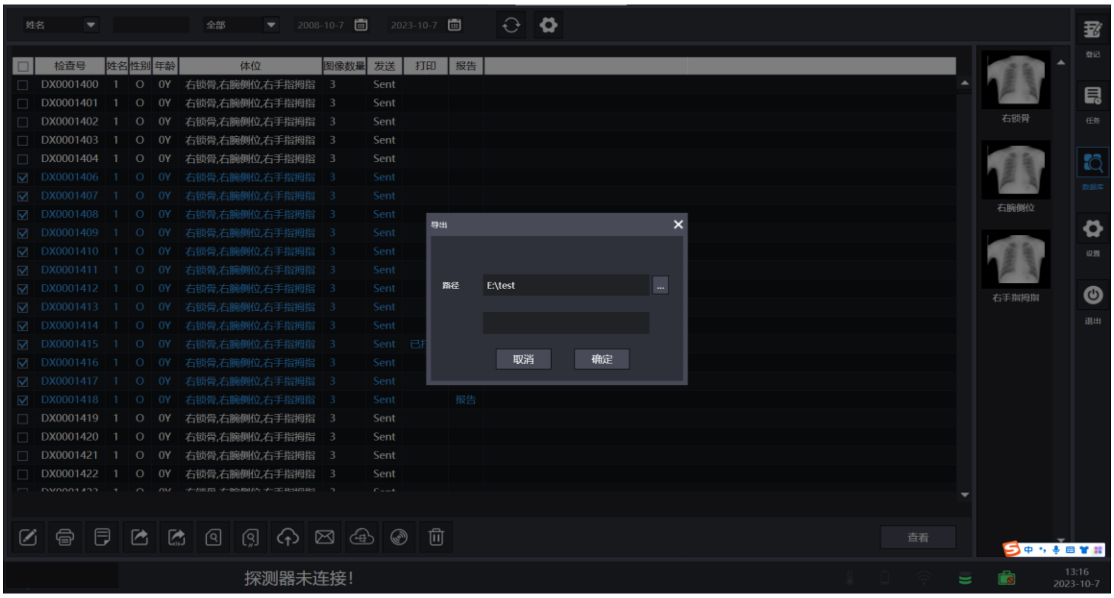


Figure 4.5-5 Excel leading-out

Select one or more messages in the database list, click the Excel Export button, and the dialog box above will pop up, select the exported path, and click the right number button to complete the export operation. The dialog box below the path shows the current export status.

Button	Function
Cancel	Return button, click it to return to the previous interface
Confirm	Save button, click it to start burning, already burned click invalid.

4.5.6 Check

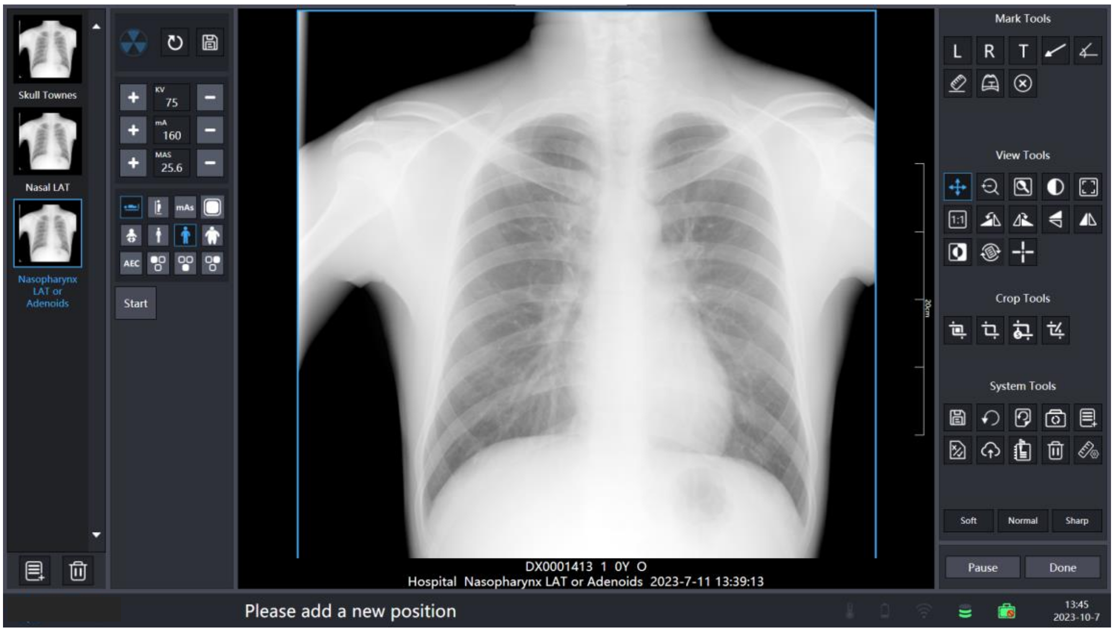


Figure 4.5-6 check

Select a case in the database list, click the review button to enter the exposure shooting interface, and you can restart the case.

4.5.7 Check NEW

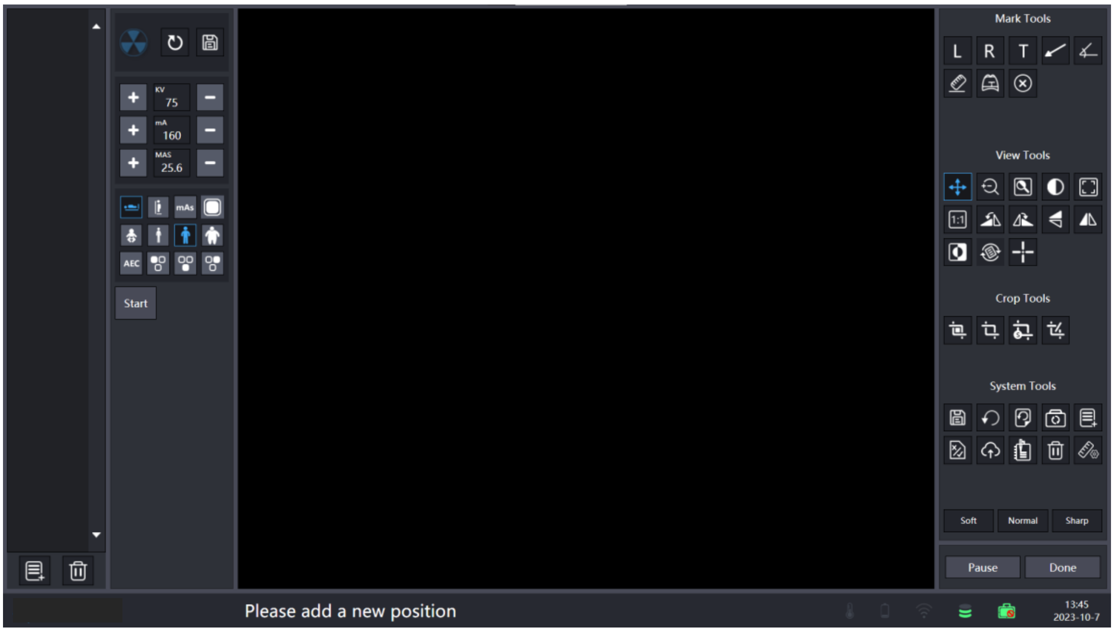


Figure 4.5-7 check NEW

Select a case in the database list and click the review button to enter the exposure shooting interface. You can operate the case again without retaining the added and taken exposure map.

4.5.8 PACS Transmission Queue

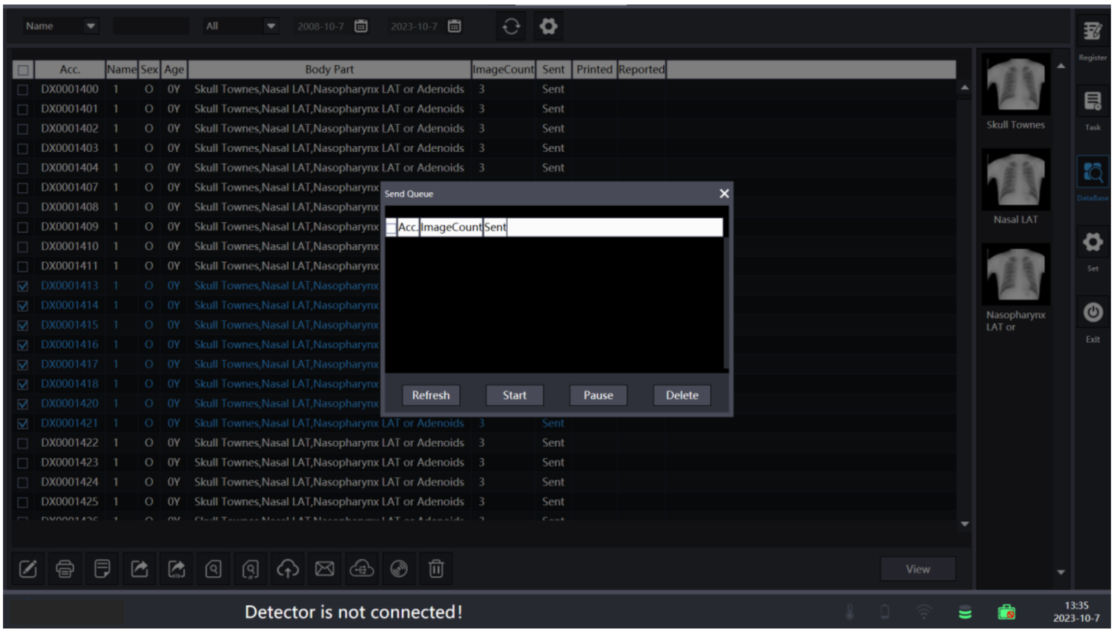


Figure 4.5-8 PACS Transmission Queue

Select local inspection, click upload button, send inspection to PACS configured in advance, as shown in the figure, the current transmission status is displayed, and the examination can be paused, started, refreshed, and deleted.

Button	Function
Refresh	Refresh the latest status.
Start	Restart the paused examination.
Pause	Pause the selected examination to stop sending.
Delete	Delete the check from sending.

4.5.9 E-mail

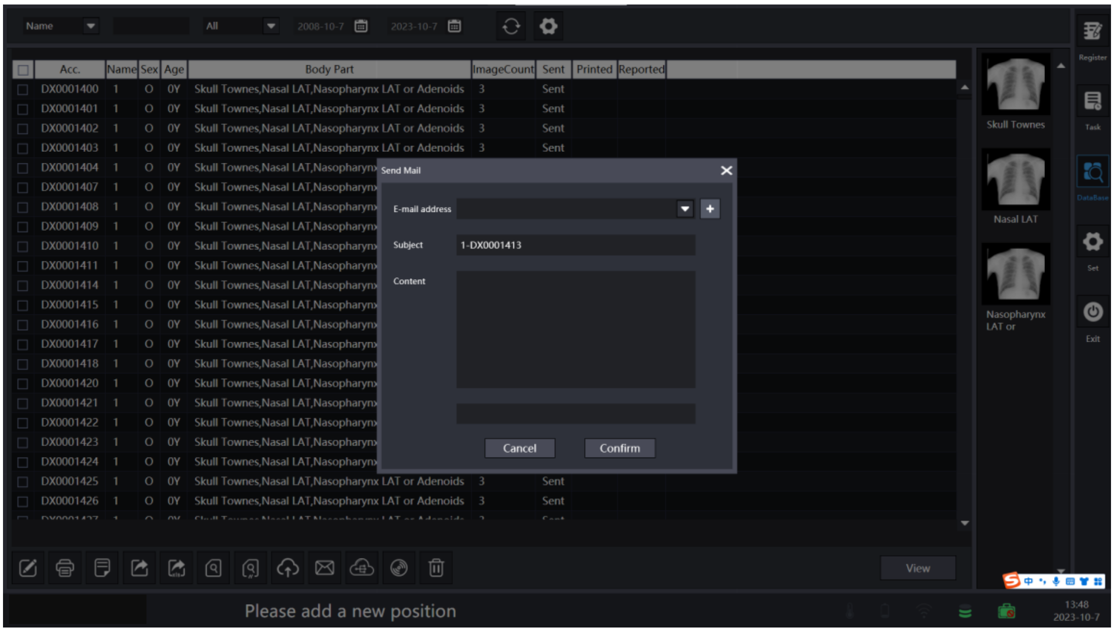


Figure 4.5-9 E-mail

Select Local Check, click the Mail button, enter the email address of the receiving user and click OK to send mail.

4.6 Configuration Tools

4.6.1

Click the Configure button and enter the password (consistent with the software password: 1) in the password dialog box which appears. As shown below:

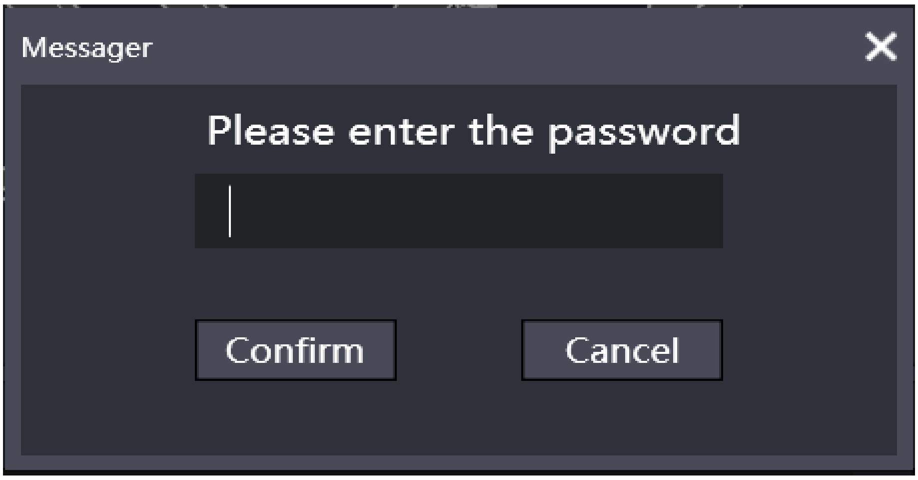


Figure 4.6-1

4.6.2 System Configuration

After entering the password, go to the configuration tool interface, as follows.

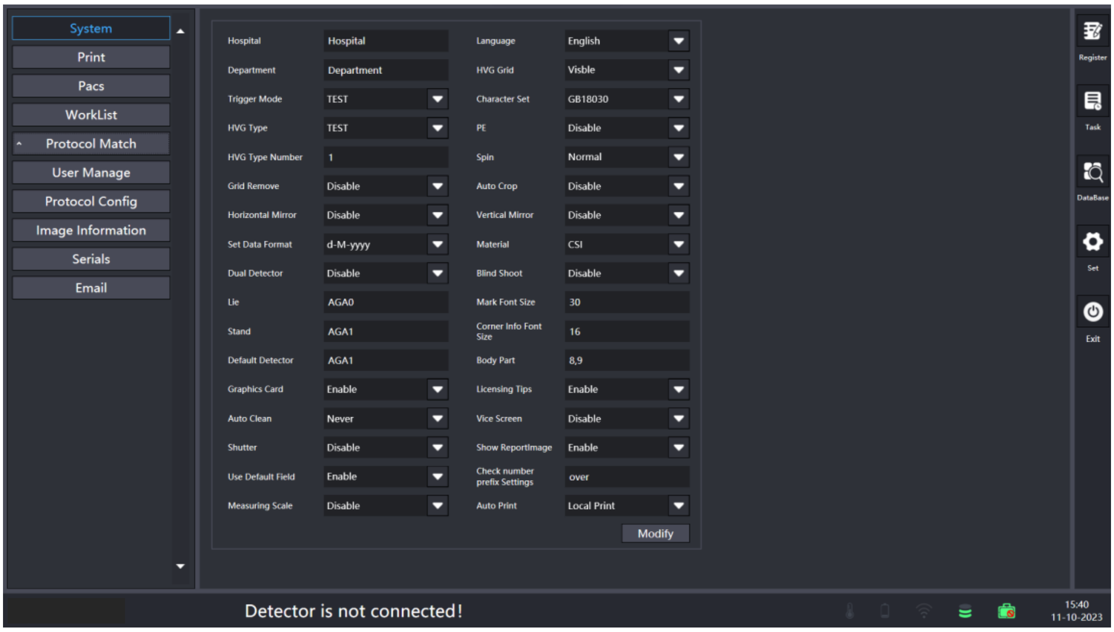


Figure 4.6-2 System Configuration

1. Configure the name of the hospital: the name of the hospital.

2. Configure the department: the name of the department.

3. Configure the exposure mode: the detector works in three modes, HST for synchronization mode, AED for exposure auto-detection mode, TEST for simulation on the map mode.

4. High voltage generator type: the default type is TEST; click the drop-down box to select the corresponding high voltage model.

5. High voltage linkage port number: according to the actual port number

configuration.

6. Grid Remove. Enable to turn on the manual remove grid function, Disable to turn off the remove grid function, Auto to open the automatic remove grid function.

7. Left and right flip: Default is not flip, Enable is flip.

8. Date format: yyyy-M-d for year-month-day d-M-yyyy for day-month-year M-d-yyyy for month-day-year.

9. Dual detector： Disable to close the double board mode, Enable to open the double board mode.

10. Lying position: fill in the lying position flat panel detector serial number.

11. Stand position: fill in the stand flat panel detector serial number.

12. Default board position: fill in the flat panel detector serial number of lying position or standing position.

13. Graphics card: part of 11 generation Intel processors because of the integrated graphics card, will lead to the exposure of the captured image cannot use the image processing function, such cases need to be set to Disable.

14. Automatic cleaning: the selected time period before the local case automatically cleared.

15. Mask: whether to zoom in after cropping to fill the browser interface.

16. Language: Simplified Chinese, Traditional Chinese, English, German, Russian, Italian, Romanian, Spanish., Portuguese, French.

17. High voltage faceplate: “Visible” shows high voltage faceplate, “Hidden” hides high voltage faceplate.

18. Byte code: Byte code of the region.

19. Body check: “Disable” not open by default, “Enable” open body check.

20. Rotate: “Normal” default not rotate, click the drop-down box to select the rotation angle to apply all body positions.

21. Auto-crop: “Disable” to turn off auto-crop, “Enable” to turn on auto-crop.

22. Up and down flip: “Disable” default not flip, “Enable” enable flip.

23. Material: select the detector material.

24. Blind shot: “Disable” not open blind shot by default, “Enable” open blind shot.

25. Marker font: marker font size: L\R\ manual marker.

26. Four corner information fonts: the font size of the four-corner information in the image.

27. Position number: 8, 9 corresponding to the two protocols displayed in the emergency room for chest radiograph orthopantomogram and chest radiograph lateral position.

28. Sub-screen: Disable to close the sub-screen, Enable to open the sub-screen

29. Authorization prompt: Disable does not open the prompt by default, Enable opens the prompt.

30. Show report image: report interface, whether to display the image to the report after selecting the position.

31. Use the default field: Disable Do not open the prompt by default, Enable open the prompt; use it with the check number prefix.

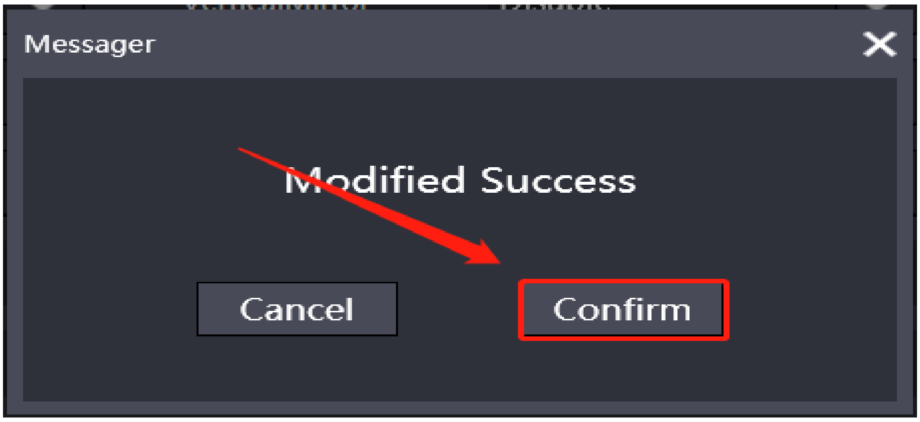
32. Check number prefix setting: custom setting check number prefix character.

33. Use the scale bar: Disable Do not open the prompt by default, and Enable will open the prompt.

34. automatic printing：Merge Print；Printer1；Printer2；Local Print

35. 

Click the modification button to prompt the software to restart to make the configuration take effect：



4.6.3 Print Configuration

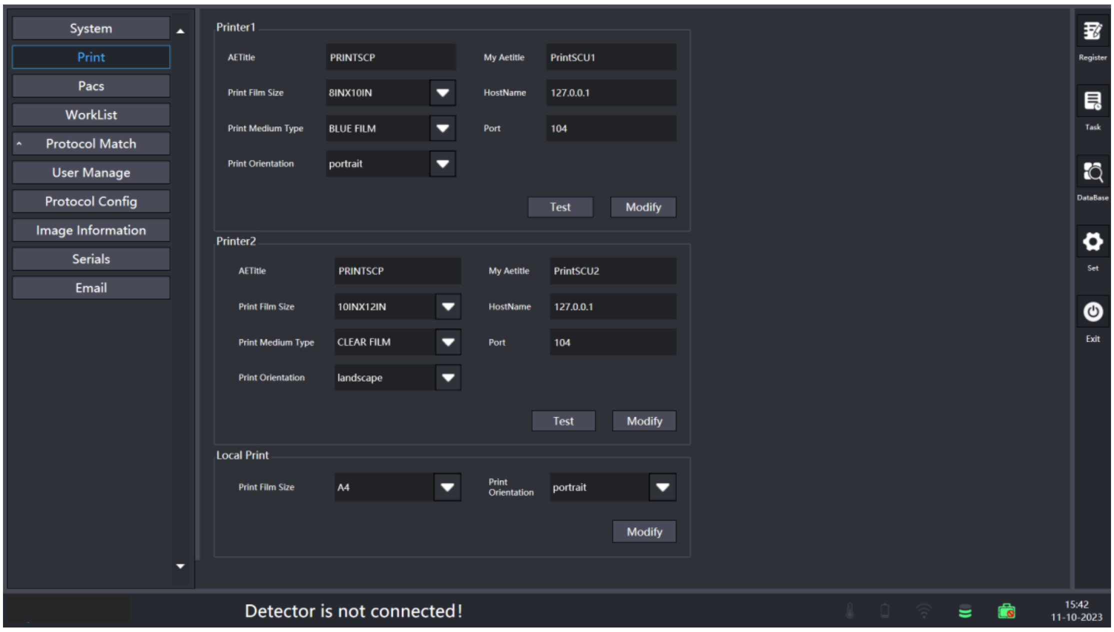


Figure 4.6-3 Print Configuration

1. AETitle: Dicom Printer AETitle.

2. Port: AETitle: AETitle name of the software.

3. Film Size: The size of the print.

4. Host name: IP address of the Dicom printer.

5. Film type: BLUE FILM, CLEAR FILM.

6. Port: The port number of the Dicom printer.

7. Print direction: Portrait for vertical printing, landscape for horizontal printing.

local printing

8. Film Size: The size of the print.

9. Print direction: Portrait for vertical printing, landscape for horizontal printing.

10. Test Click on the test to check whether the configuration is successful, Modify Click the modification button to prompt the software to restart to make the configuration take effect.

4.6.4 PACS Configuration

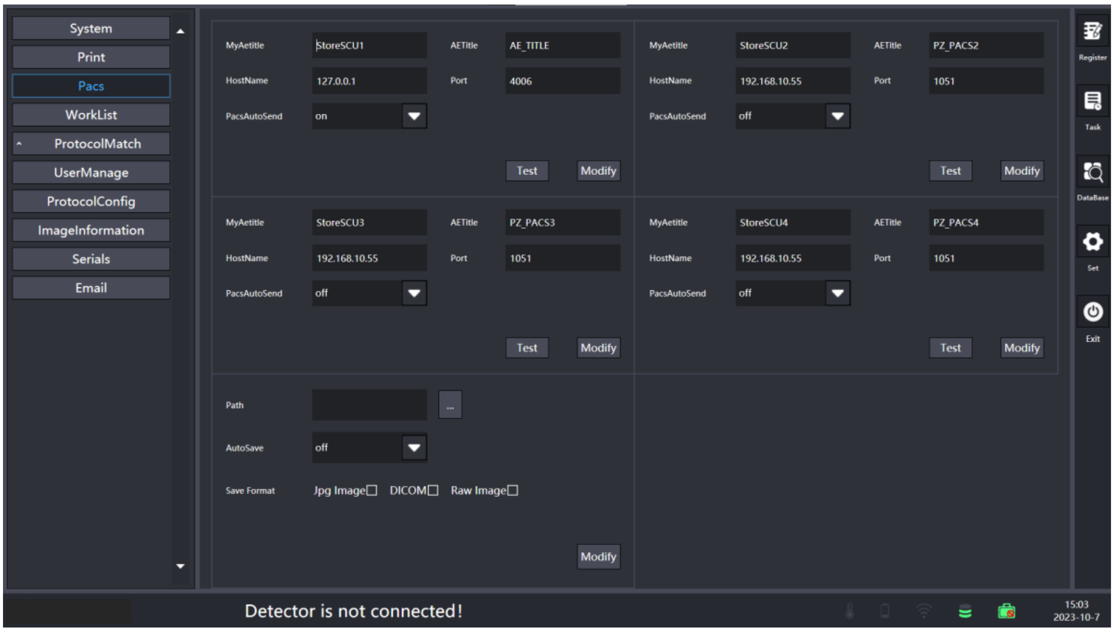


Figure 4.6-4 PACS Configuration

1. Host AETitle: AETitle name of the software.

2. AETitle： Configure the AETitle of the PACS server, consistent with the AETitle on the PACS server.

3. Hostname: Configure the IP of PACS server, consistent with the IP on PACS server.

4. Port： Configure the port number of PACS server, consistent with the port number on PACS server.

5. Auto send: “ON” is turned on, cases will be sent automatically after checking; “OFF” is turned off, cases need to be manually selected to send to PACS.

6. 

Click test to check whether the configuration is successful, 

and click modify to restart the software to make the configuration take effect.

7. Multiple PACS can be set and sent separately.

4.6.5 PASS store local locally

1. Path: The PASS storage is saved in the native path

2. Automatic storage: on is on, after the inspection, the case is automatically stored;

3. Save format: divided into jpg image, DICOM image, original image three formats

4. 

and click modify to restart the software to make the configuration take effect.

4.6.6 Work List Configuration

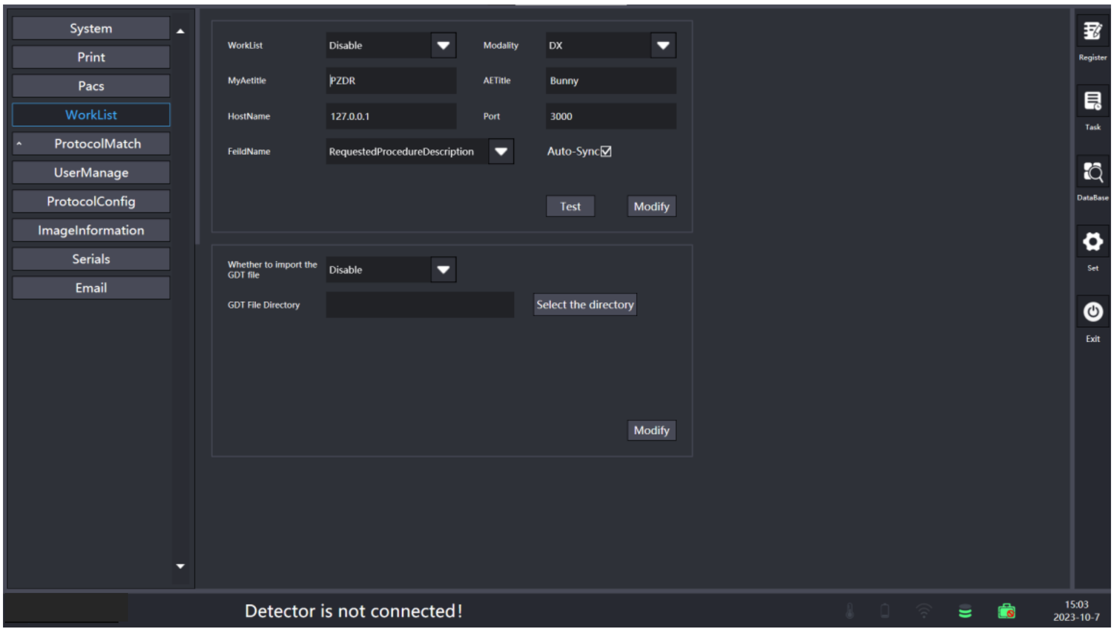


Figure 4.6-5 Worklist Configuration

1. Work List: “Enable” to turn on the connection to the Work List server, “Disable” to turn off the connection to the Work List server.
2. Device: DX or CR. In most cases, DX is used for digital x-ray type, which supports DX images. CR is used for traditional computational radiology type.
3. Host AETitle: AETitle name of the software
4. AETitle: Configure the AETitle of the WorkList server to be consistent with the AETitle on the Work List server.
5. Hostname: Configure the IP of the Work List server, consistent with the IP on the Work List server.
6. Port: Configure the port number of the Work List server, consistent with the port number on the Work List server.
7. Fields: Configure as required by the Work List server.

8. Auto Refresh: Check the box to automatically refresh the inspections sent by the Work List server, uncheck the box to not refresh.

9. 

Click Test to check whether the configuration is successful, 

click Modify to restart the software to make the configuration take effect.

4.6.7 Wordlist GDP File import automatically

1. Whether to import GDP files: Enable open, GDP files automatically sent to the work list.

2. GDP File Directory: Select the file path that you need to import into the work list.

3. 

click Modify to restart the software to make the configuration take effect.

4.6.8 Protocol Match Configuration

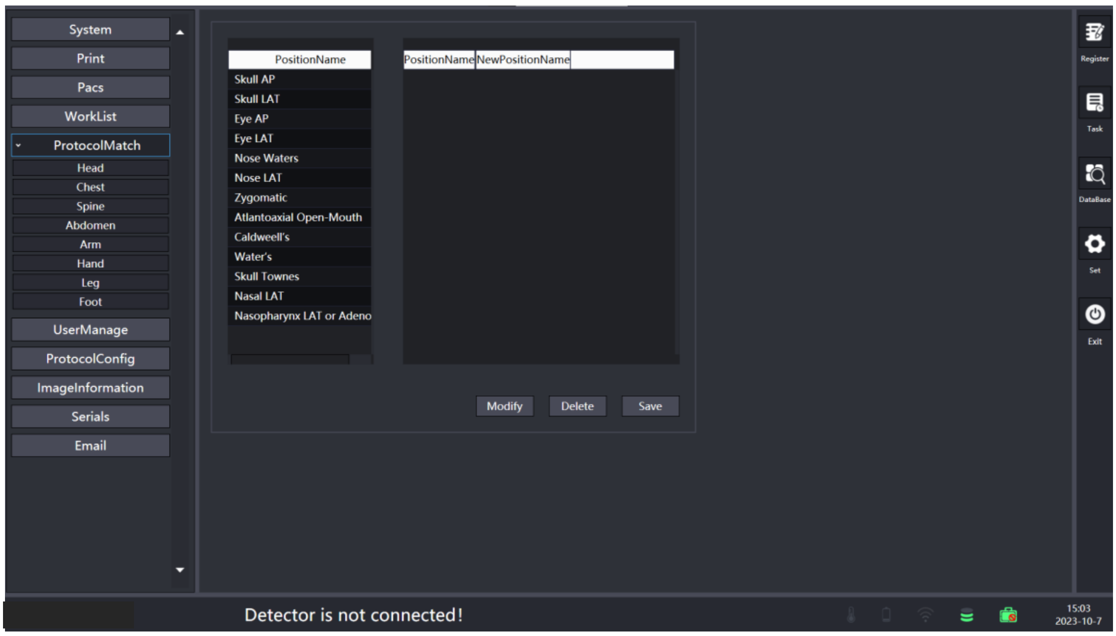


Figure 4.6-6 Protocol Match Configuration

4.6.9 User Configuration

The first list shows the local protocol name and the second list shows the protocol name passed down from Work List; select a protocol in each of the two lists and click the "Modify" button to complete the protocol matching.

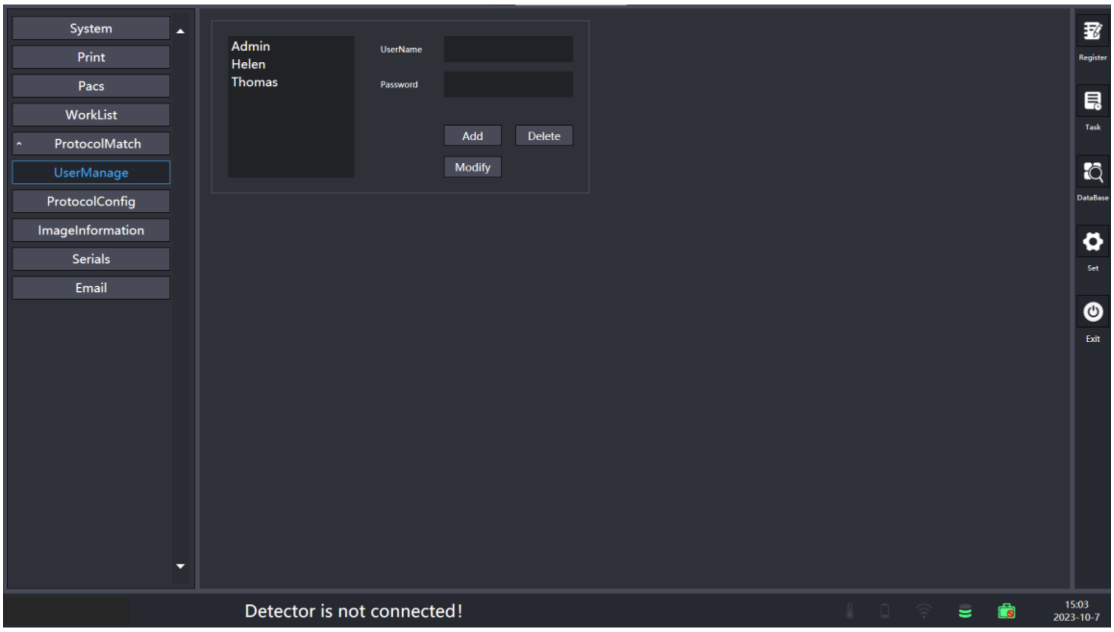


Figure 4.6-7 User Manage

The left side of the picture shows all the current users, left mouse click to view the user's name and password of the current user, the buttons below are Add User, Delete User, Modify User, click Modify and restart the software to make the configuration take effect.

(Note: Only Admin users can use the software settings bar to display the user management interface, and only Admin users can delete, add and modify user information; Admin users cannot be deleted, and only Admin users can delete patient information in the task interface and database interface).

4.6.9 Protocol Configuration

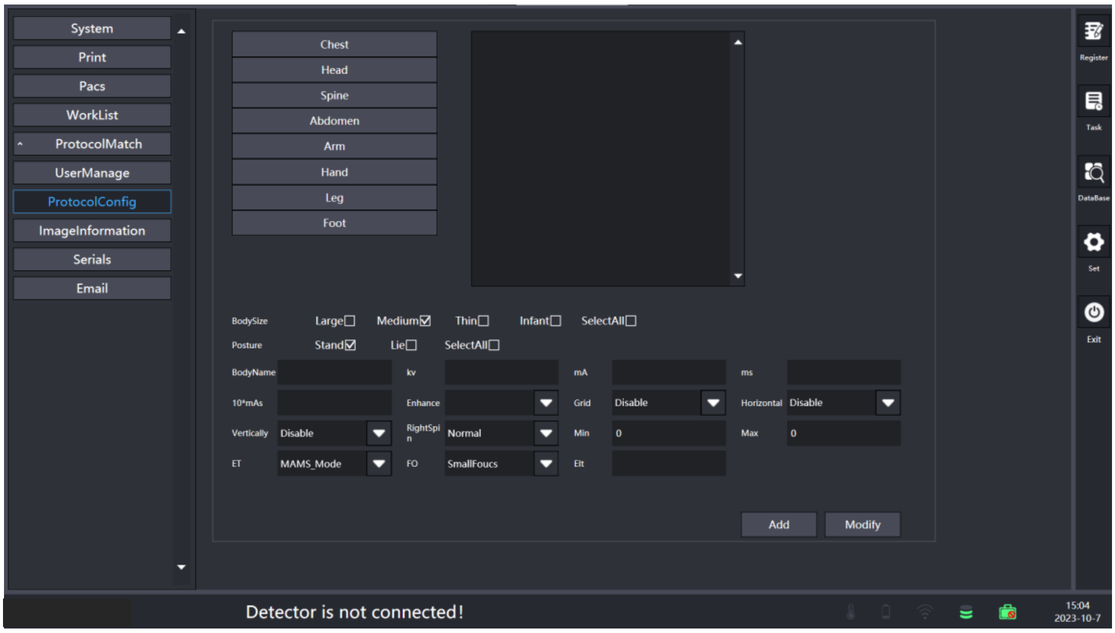


Figure 4.6-8 Protocol Config

1. Body type: choose a single or all body types

2. Posture: choose stand / lie or all postures

3. Posture name: fill in the name of the new protocol posture

4. Dose: KV/MA/MS/MAS Fill in the corresponding value to be modified

5. Enhancement: Select the post-processing file

6. Grid: “Enable” to turn on de-gridding; “Disable” to turn off de-gridding

7. Flip Left and Right: “Enable” Turn on flip; “Disable” Turn off flip

8. Flip Up and Down: “Enable” Flip On; “Disable” Flip Off

9. Rotate: Normal does not rotate, drop-down box can choose to rotate 90/180/270

10 Min/Max: set the minimum, maximum value

11. ET: Select MAS, MA/MS mode

12. FO: select large focus, small focus

13. Click 

to add new body position and its configuration, click 

to restart the software to take effect.

4.6.11 Image information

Image information is the information displayed on the exposure image and can be added or removed information distributed in the four corners of the image display by the left and right arrows (each corner has a maximum of four display information). The lower left to adjust the location of the image information display: "Corner" for the four corners, "Bottom" for the bottom center of the image.

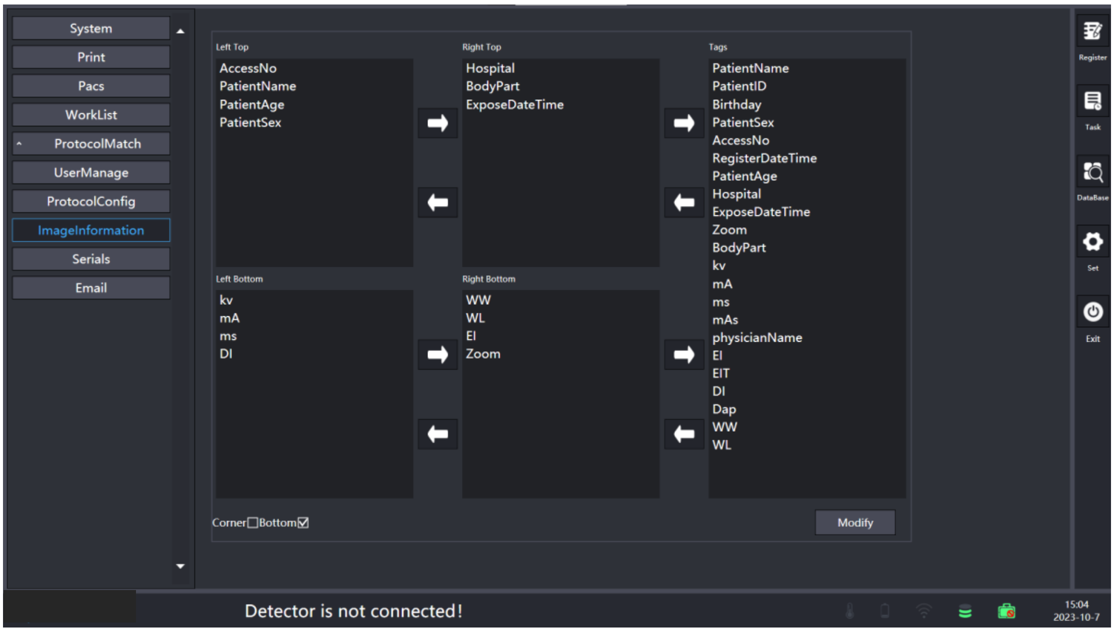


Figure 4.6-9 Image Information

4.6.12 Serials

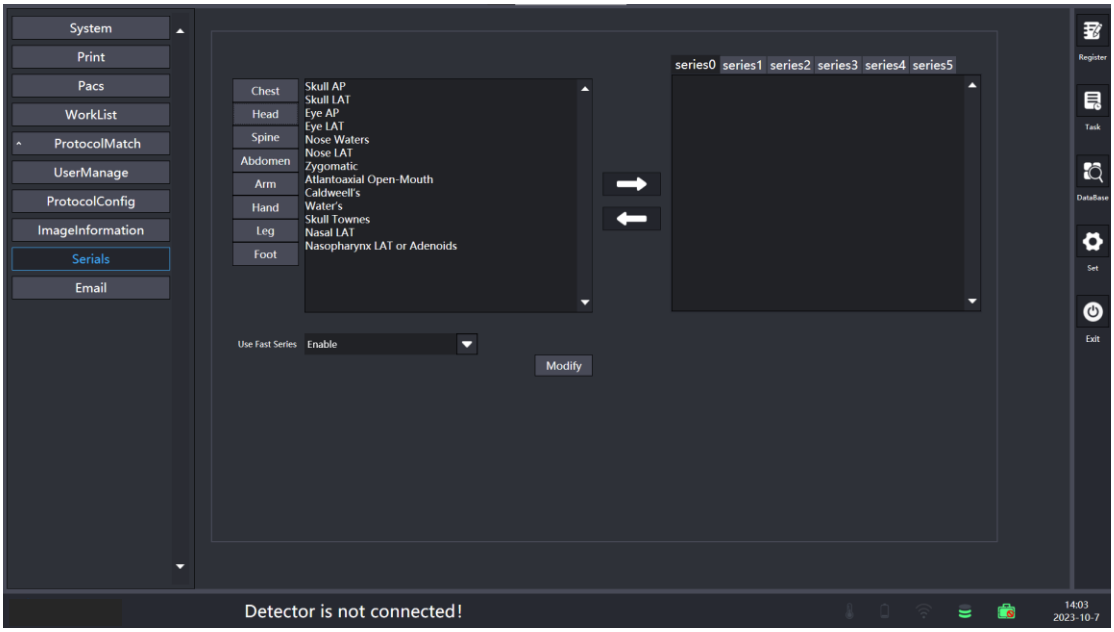


Figure 4.6-10 Serials

Provide six quick position shooting collocation schemes continuously, which should be configured in advance and set the fast sequence to Enable. When selecting the registration interface, you can select the scheme to add the part to be shot at one time, and click the arrow to add or delete the protocol in the scheme.

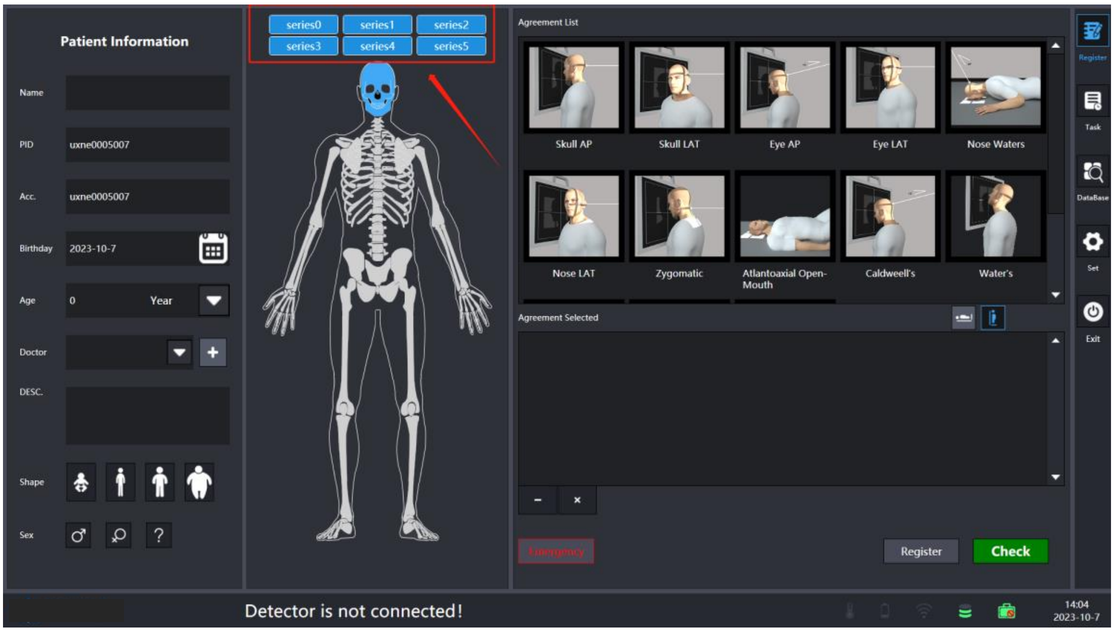


Figure 4.6-11 Registration interface

4.7 Exposure Preparation

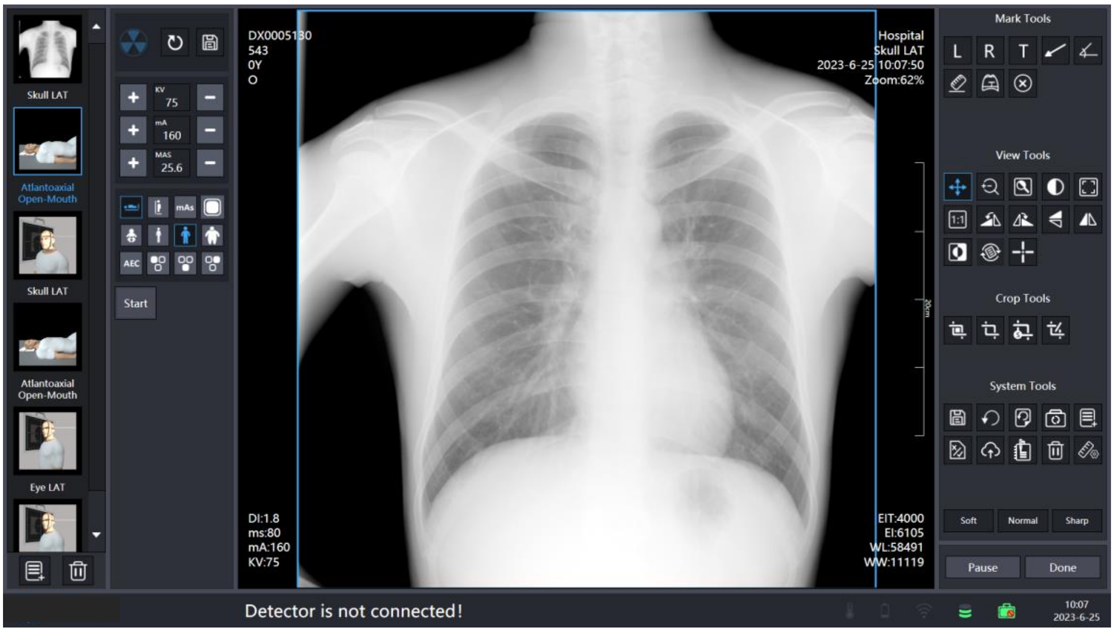
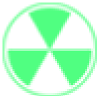
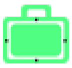


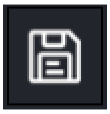
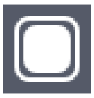
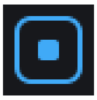
Figure 4.7-1 The Exposure Interface

The upper and lower status bar for the exposure indicator, the status instructions are as follows:

Symbol	Function
	High voltage disconnects.
	Ready to expose.
	Press the exposure handbrake first gear status, the handbrake is fully pressed at this time.
	Lights are coming out, release the handbrake in this status.
	Disconnected with the detector.
	Detector preparing.
	Detector IDLE.
	Connected with the detector and ready to expose.

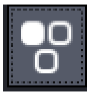
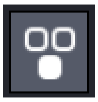
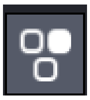
The lower status bar provides instructions for dose adjustment, high voltage mode, and settings for size focus and AEC:

Button	Function
	Plus/minus to adjust KV value.

	Plus/minus to adjust mA value.
	Plus/minus to adjust MAS value.
	In mAs mode, click to switch to mA/ms mode.
	mA/ms mode.
	Generator Reset.
	Save exposure conditions.
	Large focus, click to switch to small focus mode.
	Small focus.

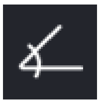
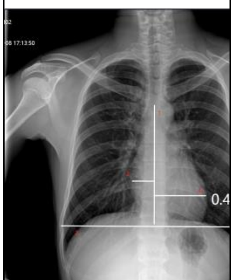
Exposure dose may be different for different patients, the left button can adjust the dose by standing and lying down, body type:

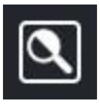
Button	**Function**
	Lying position
	Stand position
	Infant
	Lean adults

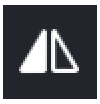
	Adults
	Fat adults
AEC	exposure time
	Left top
	inferior
	Right top
Start	Test mode, start button

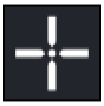
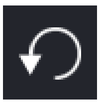
Receive the exposed image and the image can be processed with the tools on the right.

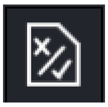
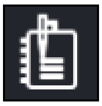
Symbol	Function	Description
L	Mark left	Clicking the button and then clicking on the image results in an L mark on the image on it. Click on the text to drag the marker
R	Mark right	Same as mark left
T	Customize tag	Same as mark left, double click on the text to enter a custom marker

	Arrows	Click on the button and click on the two coordinates on the image to determine the arrow position, the two ends of the arrow and the middle position of the arrow can be adjusted.
	Measuring angle	Click on the button and click on three coordinates on the image to determine the position of the angle measurement tool; at the same time, these three points can be dragged around to easily adjust the position of the tool.
	Measuring distance	Click on the button and click on the two coordinates to determine the position of the line, the ends of the line segment and the middle position can be adjusted.
	Cardio-thoracic ratio 	Marker 1: Take two points on the thoracic vertebrae to form the median line. Marker 2: Click on the leftmost and rightmost part of the thorax respectively to take the maximum transverse diameter of the thorax. Marker 3: mark the transverse diameter of the heart on the left and right side of the median line Cardio-thoracic ratio = (length of marker 3 + length of marker 4)/length of marker 2.

	Delete marker	Delete the selected marker
	Pan the image	Click on the button, hold down the left mouse button in the image display area and drag to pan the image up, down, left and right as desired.
	Zoom image	Click the button and hold down the left mouse button in the image display area to zoom down on the image and zoom up on the image.
	Area of interest window level and window width	Click on the button, left mouse click on the image display area, from the upper left to the lower right to draw a rectangular area (rectangular area is not displayed), after releasing the mouse, the selected area of the window that is the area of interest suitable window width window level to display the image, and the overall window width window level is different, ROI window width window level is used to optimize the local image.
	Window Level/ Window	Click this button, left-click the image display area when dragging the cursor up, the window level decreases; drag the cursor down, the

	Width	window level increases; move the cursor to the left, the window width becomes smaller; drag the cursor to the right, the window width increases.
	Adaptable button	Click this button then current image will zoom in or out automatically to fill all image area window.
	Real size	Click this button to display image in an actual size.
	Rotate 90 degrees counterclockwise	Click this button to rotate the image 90 degrees counterclockwise.
	Rotate 90 degrees clockwise	Click this button to rotate the image 90 degrees clockwise.
	Flip up and down	Click this button to flip the image vertically and swap it up and down.
	Flip left and right	Click this button to flip the image vertically and swap left and right.

	Inverse	Invert the image color.
	Free rotation	Click the Rotate button, fill in the angle number, and click Confirm to rotate the image.
	Grayscale values	Move the cursor over the image to see the grayscale value of the pixel point.
	Image crop: as desired crop out rectangular image areas	Click on the button, a rectangle box will appear in the image display area, drag the rectangle box and adjust the size of the rectangle box to select the area to be cropped.
	Save crop	Save the image in the rectangular box area and the image will be automatically scaled to the entire image area when finished.
	Undo crop	Undo the crop, the image is restored to the original image size and automatically scaled to the entire image area.
	Save marker	Add marks, arrows, lines, angles into the image.
	Reset	Click this button to restore the image to its original state.

	Reprocessing	Reprocess the image.
	Reshoot	Save body position, recollect the image.
	Add protocol	Add new protocols.
	Reject picture / cancel rejection	Click the button to expose the image to become unavailable and add a new position, click again to restore the available state
	upload	Send single image to PACS.
	Measurement Correction	Mark two points to determine the length of the line segment, enter the exact length of the line segment, click OK, correction is complete.
	Update exposure parameters	Modify exposure parameters.
	Delete	Delete image
Soft	Soft	Adjust the image transitions to make the transition round and soft.
Normal	Normal	Intermediate between soft and sharp.

	Sharp	Sharpens the edges where lines or black and white images meet in sudden changes.

The two buttons at the bottom right, used in the completion of the exposure, click to pause then exit the exposure, back to the registration interface, the information will be saved to the task list, within the task list can be re-entered to check the new protocol exposure; click to complete and exit the exposure, the information saved to the database list cannot be added again after the protocol to continue exposure.

Button	**Function**
	Pause inspection and return to the registration screen (information is saved to the task list).
	Complete inspection and return to the registration screen (information is saved to the database list).

4.8 Review Interface

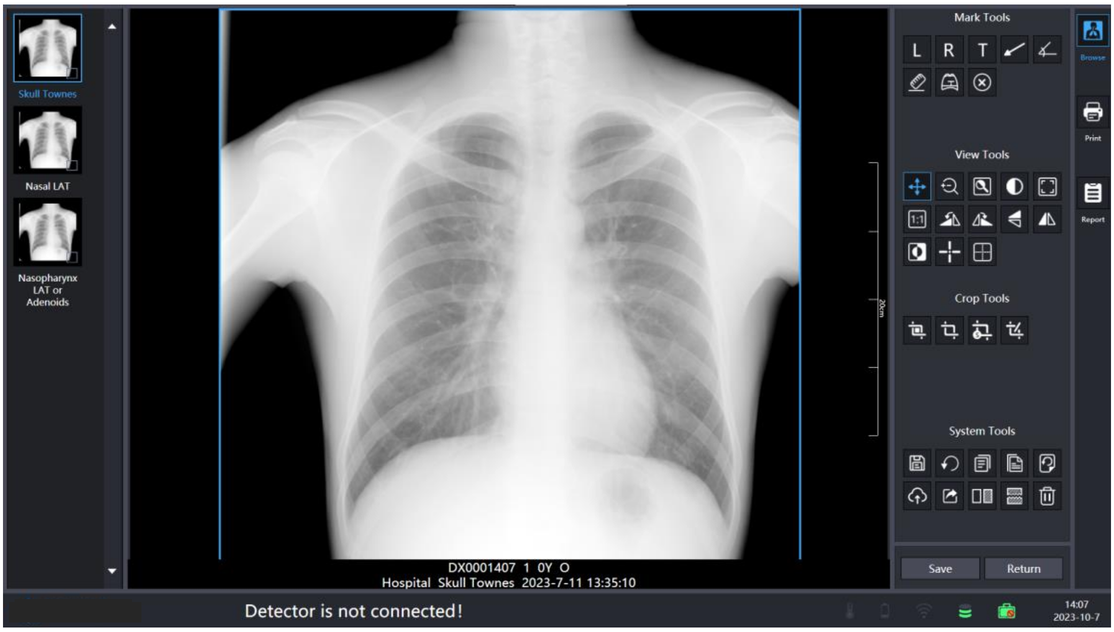
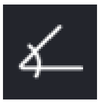
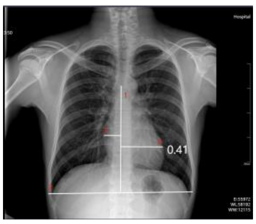
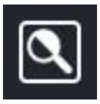


Figure 4.8-1 Review Interface

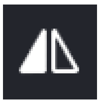
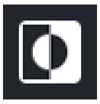
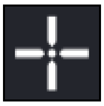
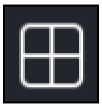
After checking the completed images in the database, enter the browse interface and use the tools for simple processing of the images.

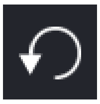
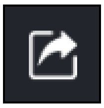
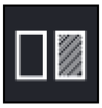
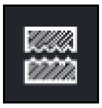
Symbol	Function	Description
L	Mark left	Click the button and then click the image to have the L marker on the image. Click on the text to drag the marker.
R	Mark right	Same as left marker.
T	Customize tag	Click the button and then click on the image and double click on the text to enter a custom

		marker.
	Arrows	Click on the button and click on two coordinates on the image to determine the arrow position, the arrow position can be adjusted at both ends and in the middle of the arrow.
	Measuring angle	Click on the button and click on three coordinates on the image to determine the position of the angle measurement tool; the three points can be dragged at the same time to adjust the position of the tool easily.
	Measuring distance	Click on the button and click on the two coordinates to determine the position of the line, the ends of the line segment and the middle position can be adjusted.
	CTR 	Marker 1: taking two points on the thoracic vertebrae to form the median line. Marker 2: take the maximum transverse diameter of the thorax by clicking on the leftmost and rightmost part of the thorax respectively.

		Marker 3: mark the transverse diameter of the heart on the left and right sides of the median line. Cardio-thoracic ratio = (length of marker 3 + length of marker 4)/length of marker 2.
	Delete mark	Delete the selected marker.
	Pan the image	Click on the button, hold down the left mouse button in the image display area and drag to pan the image up, down, left and right as desired.
	Zoom image	Click the button and hold down the left mouse button in the image display area to zoom down on the image and zoom up on the image.
	Area of interest window level and window width	Click on the button, left mouse click on the image display area, from the upper left to the lower right to draw a rectangular area (rectangular area is not displayed), after releasing the mouse, the selected area of the window that is the area of interest suitable

		window width window level to display the image, and the overall window width window level is different, ROI window width window level is used to optimize the local image.
	Window level / window width	Click the button, left-click the image display area when dragging the cursor up, the window level decreases; drag the cursor down, the window level increases; move the cursor to the left, the window width becomes smaller; drag the cursor to the right, the window width increases.
	Adaptable button	Click on this button and the window is automatically scaled so that the image is displayed fully and maximally in the window.
	Real size	Click this button to restore the image to its actual size.
	Rotate 90 degrees counterclockwise	Clicking this button rotates the image 90 degrees counterclockwise.
	Rotate 90 degrees clockwise	Clicking this button rotates the image 90 degrees clockwise.
	Flip up and down	Click this button to flip the image vertically and swap it up and down.

	Flip left and right	Click this button to flip the image horizontally and swap it left and right.
	Inverse	Invert the image color.
	Grayscale values	Move the cursor over the image to see the grayscale value of the pixel.
	Browsing mode	Selects the window mode of the browsing interface.
	Image crop: as desired crop out rectangular image areas	Click on the button, a rectangle box will appear in the image display area, drag the rectangle box and adjust the size of the rectangle box to select the area to be cropped.
	Save crop	Save the image in the rectangular box area and the image will be automatically scaled to the entire image area when finished.
	Undo crop	Undo the crop, the image is restored to the original image size and automatically scaled to the entire image area.
	Save marker	Add marks, arrows, lines, angles into the image.

	Reset	Click this button to restore the image to its original state.
	Copy	Copy the image information of the selected body position into the pasteboard.
	Paste	Copy the information in the pasteboard to the current inspection.
	Reprocessing	Enter the reprocessing interface.
	Upload	Send single image to PACS.
	Export	Export selected images to a specified location.
	Comparison	One or two images can be selected for comparison.
	Stitching	Stitch the two images together.
	Delete	Delete the selected image.

Click on any of the two icons at the bottom right to return to the completion list after viewing.

Button	**Function**

	Save current changes.
	Return to database list.

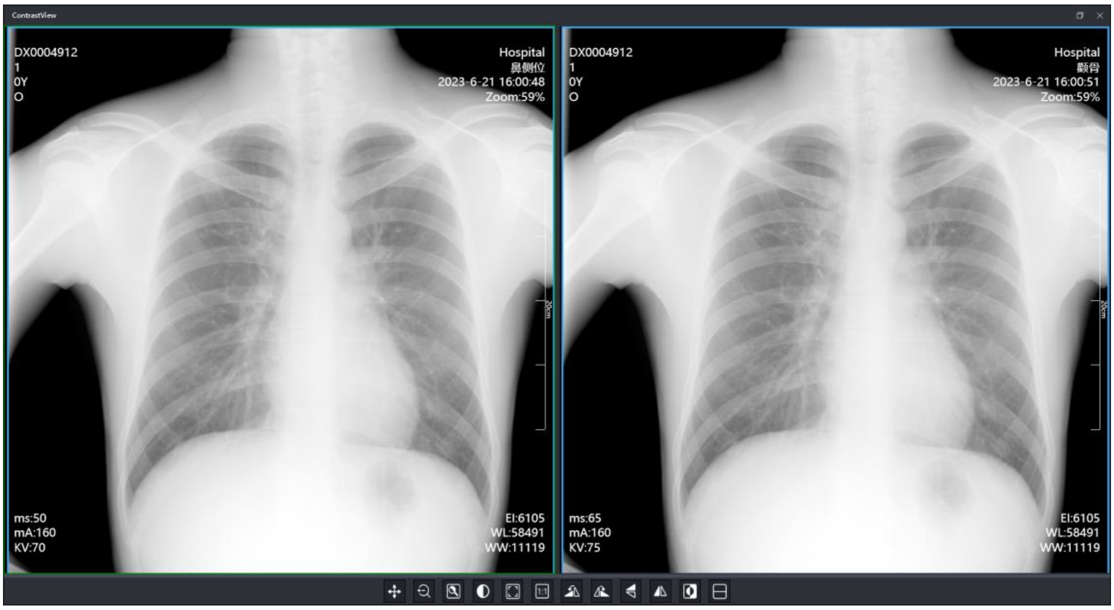
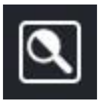
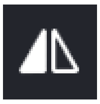
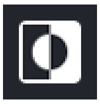
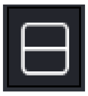
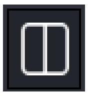


Figure 4.8-2 Interface comparison

	Pan the image	Click this button, hold down the left mouse button in the image display area and drag, the image will be panned up, down, left and right as needed.
	Zoom image	Click this button and hold down the left click of the mouse in the image display area to zoom down on the image and zoom up on the image.

	Area of interest window level and window width	Click on the button, left mouse click on the image display area, from the upper left to the lower right to draw a rectangular area (rectangular area is not displayed), after releasing the mouse, the selected area of the window that is the area of interest suitable window width window level to display the image, and the overall window width window level is different, ROI window width window level is used to optimize the local image.
	Window level / window width	Click this button, left-click the image display area when dragging the cursor up, the window level decreases; drag the cursor down, the window level increases; move the cursor to the left, the window width becomes smaller; drag the cursor to the right, the window width increases.
	Adaptable button	Click this button then current image will zoom in or out automatically to fill all image area window.
	Real size	Click this button to display image in an

		actual size.
	Rotate 90 degrees counterclockwise	Click this button to rotate the image 90 degrees counterclockwise.
	Rotate 90 degrees clockwise	Click this button to rotate the image 90 degrees clockwise.
	Flip up and down	Click this button to flip the image vertically and swap it up and down.
	Flip left and right	Click this button to flip the image horizontally and swap it left and right.
	Inverse	Invert the image color.
	Vertical layout	Two images side-by-side layout.
	Horizontal layout	Two images side-by-side layout.

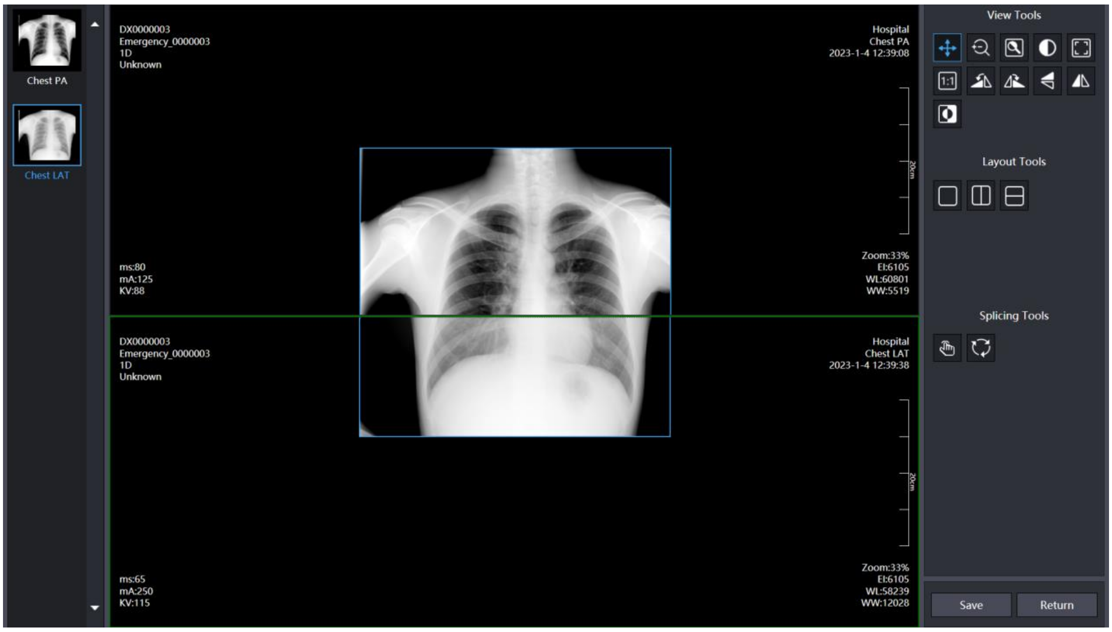
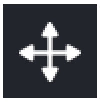
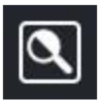
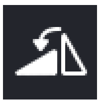
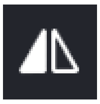
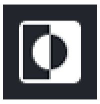


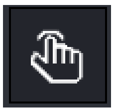
Figure 4.8-3 stitching interface

	Pan the image	Click this button, hold down the left mouse button in the image display area and drag, the image will be panned up, down, left and right as needed.
	Zoom image	Click this button and hold down the left click of the mouse in the image display area to zoom down on the image and zoom up on the image.
	Area of interest window level and window width	Click on the button, left mouse click on the image display area, from the upper left to the lower right to draw a rectangular area (rectangular area is not displayed), after

		releasing the mouse, the selected area of the window that is the area of interest suitable window width window level to display the image, and the overall window width window level is different, ROI window width window level is used to optimize the local image.
	Window Level/ Window Width	Click this button, left-click the image display area when dragging the cursor up, the window level decreases; drag the cursor down, the window level increases; move the cursor to the left, the window width becomes smaller; drag the cursor to the right, the window width increases.
	Adaptable button	Click this button then current image will zoom in or out automatically to fill all image area window.
	Real size	Click this button to display image in an actual size.
	Rotate 90 degrees counterclockwise	Click this button to rotate the image 90 degrees counterclockwise.

	Rotate 90 degrees clockwise	Click this button to rotate the image 90 degrees clockwise.
	Flip up and down	Click this button to flip the image vertically and swap it up and down.
	Flip left and right	Click this button to flip the image horizontally and swap it left and right.
	Inverse	Inverse the image color.

The Layout toolbar, with multiple layout functions, allows two images to be stitched together into a single image by using the Stitch tool.

Symbol	Function
	Single image layout
	Two images side-by-side layout
	Two images side-by-side layout
	Stitch two images together in a preset position
	Stitching automatically

4.9 Print List

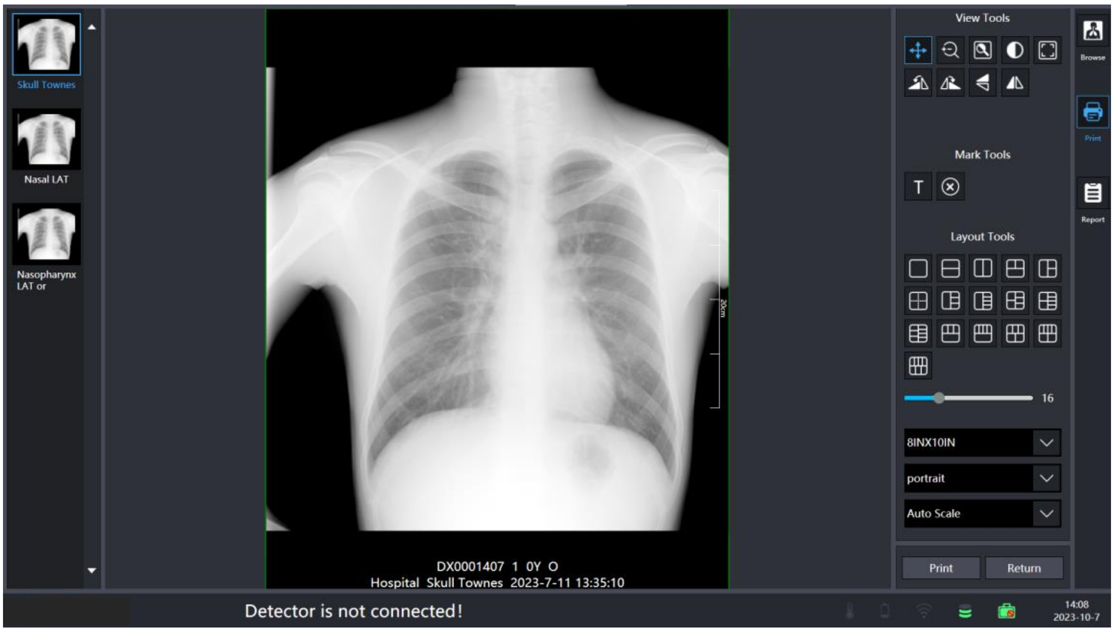
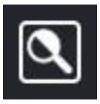
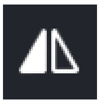
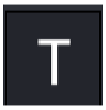


Figure 4.9-1 Print List

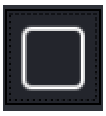
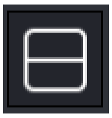
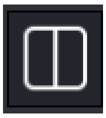
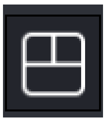
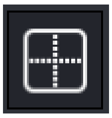
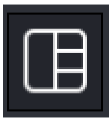
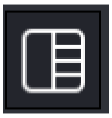
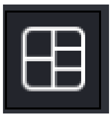
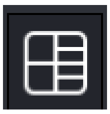
Simple manipulation of the image with the right tool before printing.

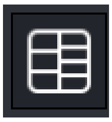
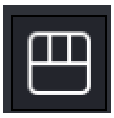
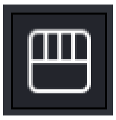
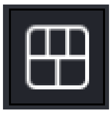
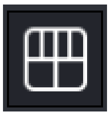
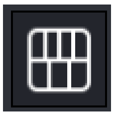
Symbol	Function	Description
	Pan the image	Click this button, hold down the left mouse button in the image display area and drag, the image will be panned up, down, left and right as needed.
	Zoom image	Click this button and hold down the left click of the mouse in the image display area to zoom down on the image and zoom up on

		the image.
	Area of interest window level and window width	Click on the button, left mouse click on the image display area, from the upper left to the lower right to draw a rectangular area (rectangular area is not displayed), after releasing the mouse, the selected area of the window that is the area of interest suitable window width window level to display the image, and the overall window width window level is different, ROI window width window level is used to optimize the local image.
	Window Level/ Window Width	Click this button, left-click the image display area when dragging the cursor up, the window level decreases; drag the cursor down, the window level increases; move the cursor to the left, the window width becomes smaller; drag the cursor to the right, the window width increases.
	Adaptable button	Click this button then current image will zoom in or out automatically to fill all image

		area window.
	Real size	Click this button to display image in an actual size.
	Rotate 90 degrees counterclockwise	Click this button to rotate the image 90 degrees counterclockwise.
	Rotate 90 degrees clockwise	Click this button to rotate the image 90 degrees clockwise.
	Flip up and down	Click this button to flip the image vertically and swap it up and down.
	Flip left and right	Click this button to flip the image horizontally and swap it left and right.
	Customize tag	Click the button and then click on the image and double click on the text to enter a custom marker.
	Delete	Select the marker on the image and click to delete it.

Layout toolbar with multiple layout functions to stitch multiple images onto a single film for printing. Drag rule: Left-click to select an image and drag it to the target layout box.

	Individual image layout
	Two images with a late ral layout
	Two images are laid layout
	Three images are juxtaposed on and down in the horizontal layout
	Three images are placed together
	Custom layout
	Four images are placed together
	Five images are placed together
	Five images are placed together
	Six images are placed together

	Seven images are laid parallel
	four images are juxtaposed on and under the horizontal layout
	Five images are juxtaposed on and under the horizontal layout
	Five images are laid parallel
	Six images are placed together
	Seven images are laid parallel
	Adjust the corner font size

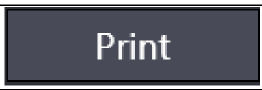
Film size selection and print direction tool.

	Click the drop-down box to select print film size.

	Click the drop-down box to choose the direction of printing.

Click the first button at the bottom right to print and the second button to exit printing.

Local printing or Dicom printing depends on the settings in the configuration.

Button	**Function**
	Print button, click it to print film
	Return button, click it and return to the database.

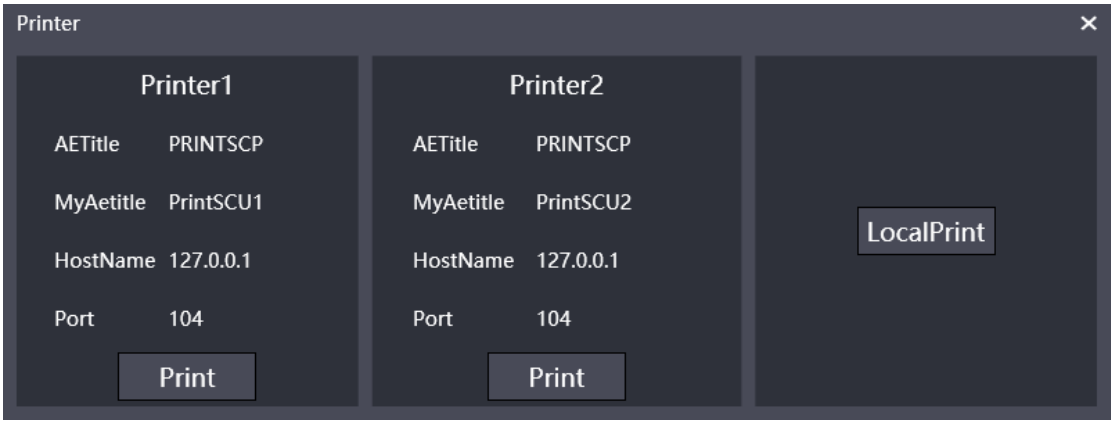


Figure 4.9-2 Printer Selection Interface

4.10 Diagnostic Report

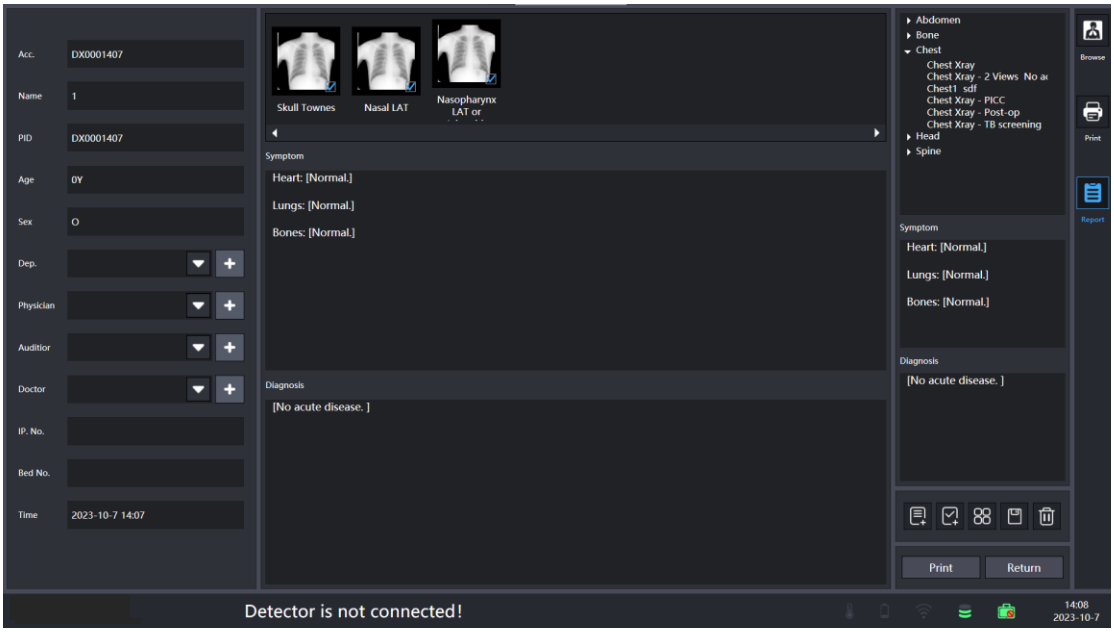
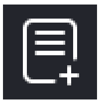
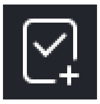
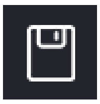


Figure 4.10-1 Diagnostic Report

Completing the addition of disease descriptions, diagnoses based on images, the descriptions and diagnosis are saved to the case.

Symbol	Function
	Add description
	Add diagnosis
	Add description and diagnosis
	Save changes to the database

	Clear description and diagnosis
Return	Click and return to the database
Print	Click it to enter print interface.

4.11 Report Template

In the PZDR installation catalog, ReportEditTool.exe is the report template editing tool, and ReportDocumentTemplate.xaml under the Configs\ReportTemplate path is the initial report template.

After opening the editing tool, click "Model" -> "OpenModelFile" on the toolbar to select the initial report template to edit, and right-click on the blank area to select "Hide Gridlines" or "Displays Gridlines" to hide or show the background grid (to assist in adjusting the cells), and select "Change the background Color" to change the template Background color.

Click "Controls"->"Image", then click on the blank area to add a picture, double-click on the picture to select a local picture, overlay added to the report template. Click "Controls"->"TextBlock", then click the blank area to add a text box, double-click it to edit the text content. The "Line" button can add a line, which can be edited separately.

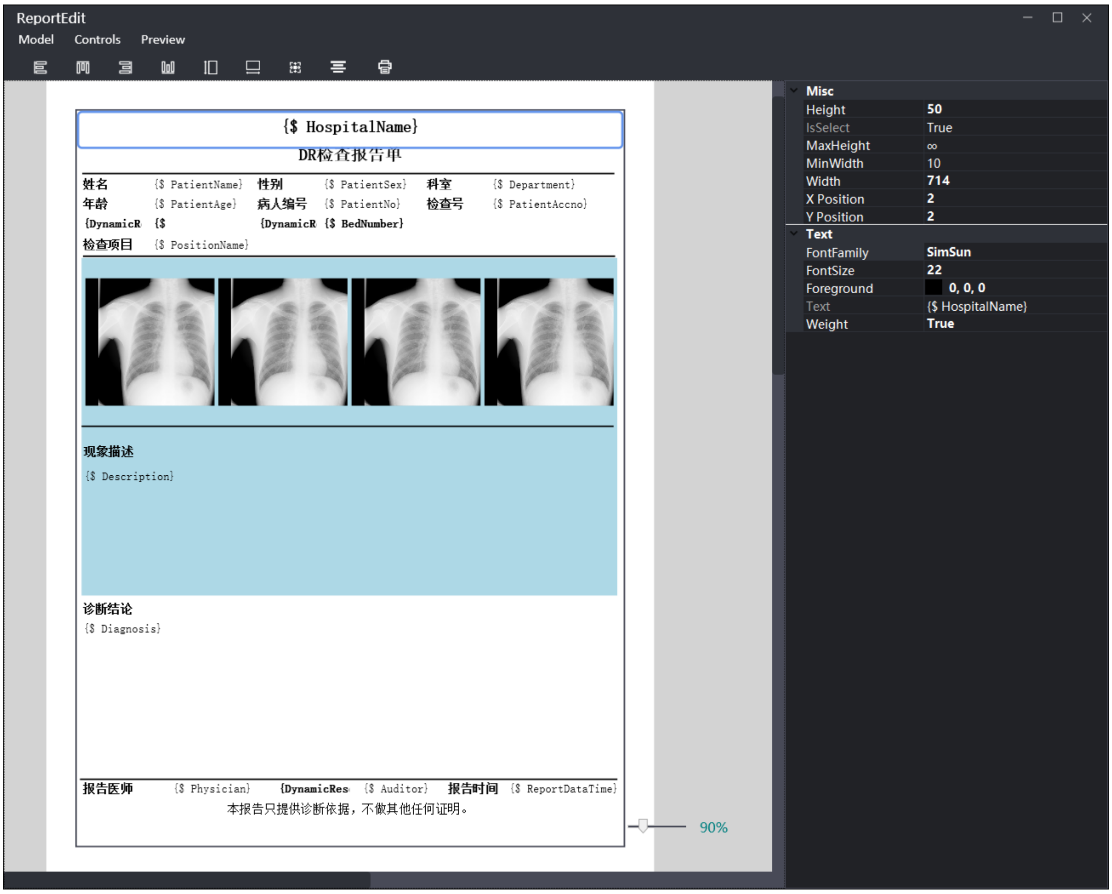
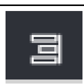
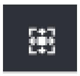
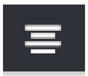
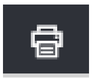


Figure 4.11-1 Report Template Editing Interface

The keyboard left "Ctrl" and the left mouse button can be used together to select multiple display units and use the upper toolbar for layout.

Symbol	**Function**
	Selected cell left-aligned
	Selected cell top-aligned
	Selected cell right-aligned
	Selected cell bottom-aligned
	Selected cells aligned top and bottom
	Selected cell left-right aligned

	Selected cell size aligned
	Selected cell center aligned
	Report Template Print Preview

When a single display cell is selected, use the right toolbar to change the properties of the cell and its fonts.

Misc: Properties of the display cell

Height: Height

MaxHeight: maximum height

MinWidth: minimum width

Width: width

X Position: the horizontal coordinate of the upper left corner

Y Position: the vertical coordinate of the upper left corner

Text: Properties of the text

FontFamily: font style

FontSize: font size

Foreground: font color

Text: (please do not change)

4.12 Reprocessing

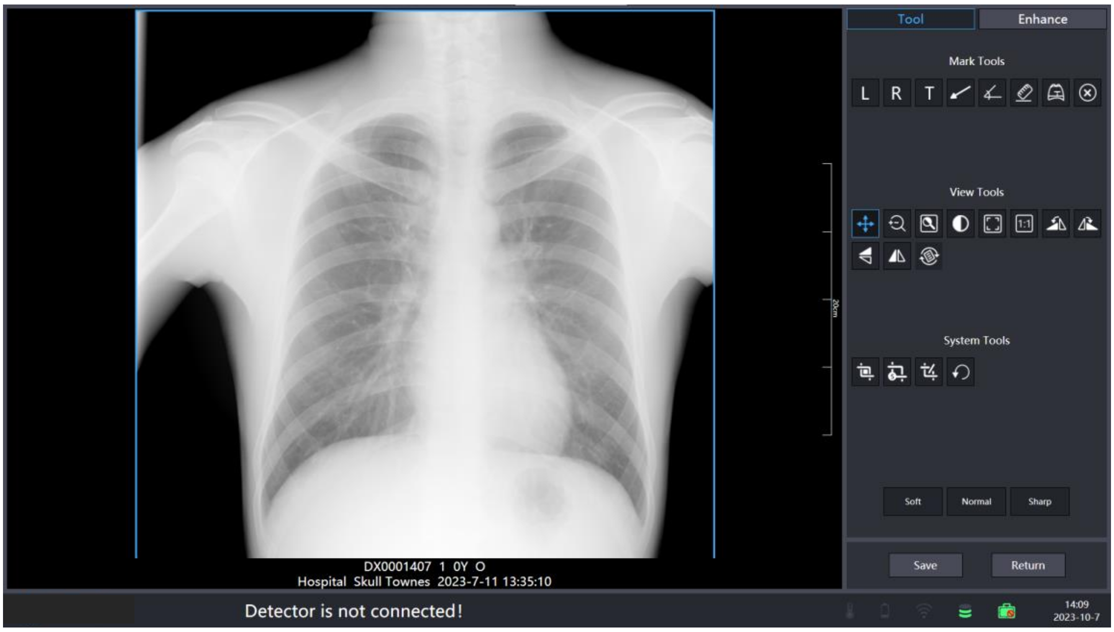


Figure 4. 12-1 Reprocessing Interface

The toolbar has the same functions as the acquisition interface, and the image enhancement column has the image enhancement parameters of contrast, edge enhancement, noise suppression, and gamma correction, respectively.

Symbol	Description
Save	Save the current parameters to the current body position post-processing file.
Process	Process the image by current parameters.
Reset	Undo the current processing effect.

Save	Do not save the current processing parameters, save the current processing effect and return to the previous interface.
Return	No saving of processing parameters, no saving of processing effects, return to the previous interface.

4.13 Offline Mode

Enter the settings, set the blind shot to "Enable", confirm the changes and restart the software.

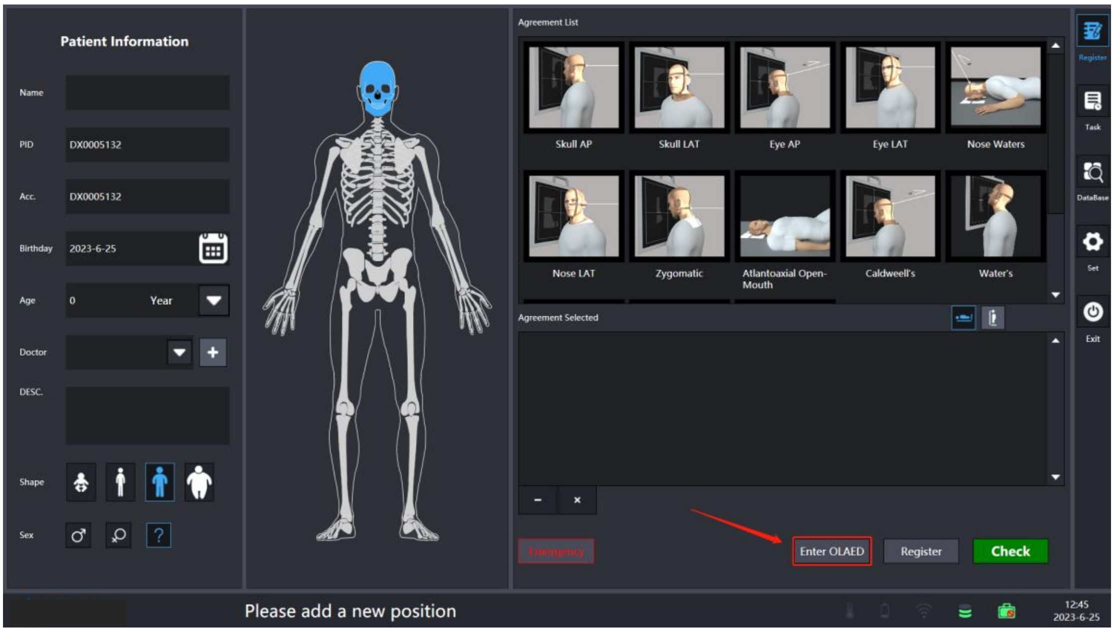


Figure 4. 13-1

Click to enter the offline mode, create a case, and make an exposure.

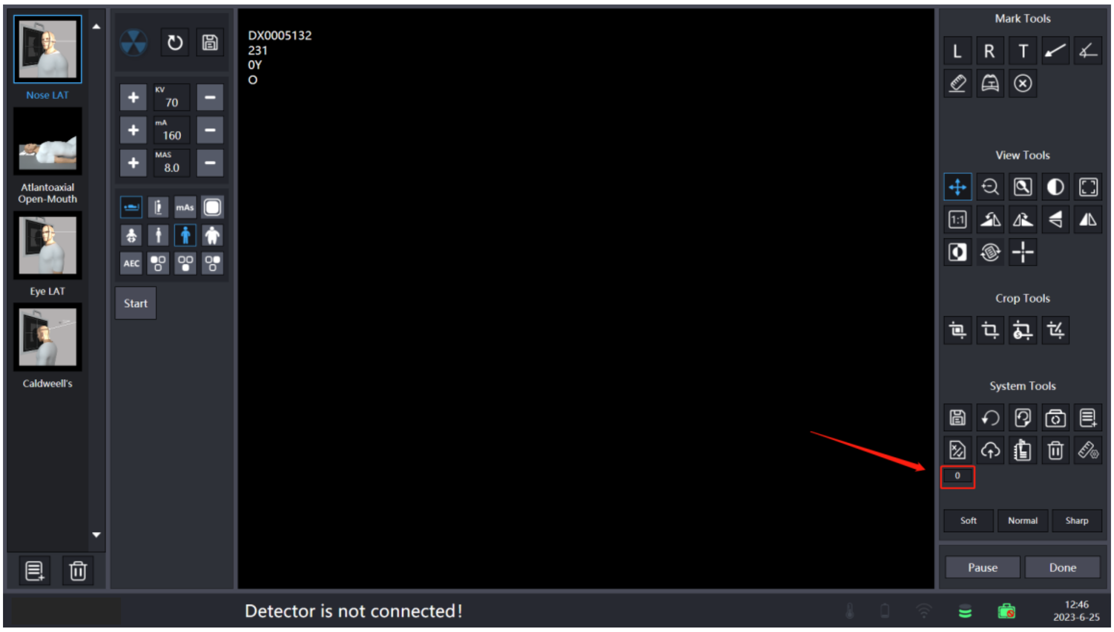


Figure 4. 13-2

Access the case and click on the red box to take out the image. Return to the registration interface to exit the blind shoot mode.

4.14 Dual-detector Setting and Using

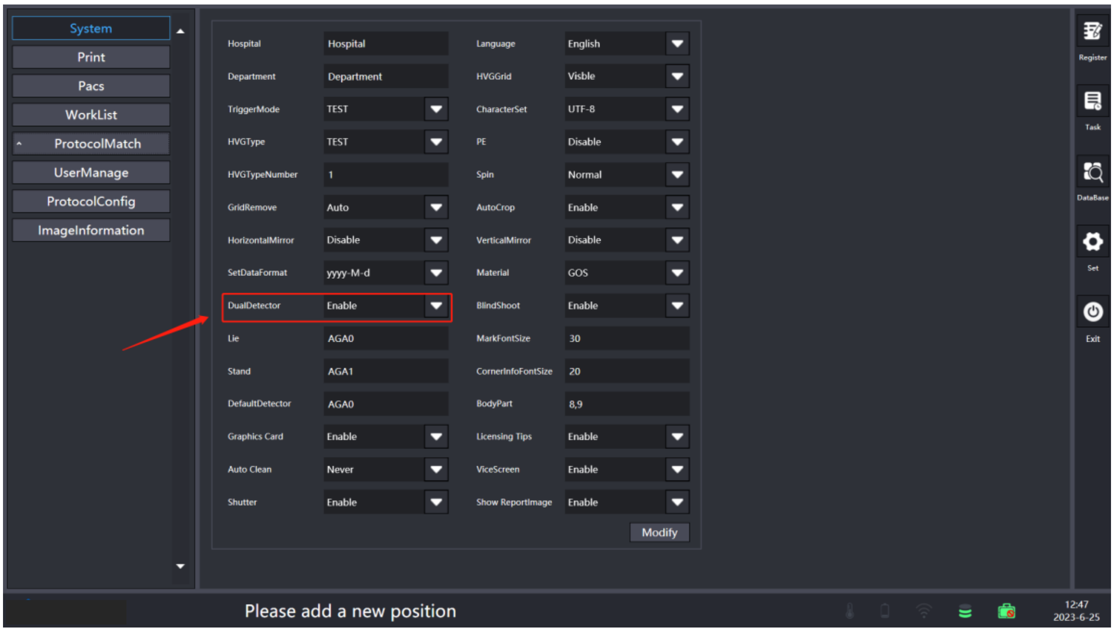


Figure 4. 14-1 System Interface

The remaining functional settings are the same as the single board, set the dual panel, the dual panel settings from "Disable" to "Enable", the lying position is written to a detector serial number, the standing position is written to another flat serial number, the default board position is written to one of the serial numbers Click on the "Disable" button to confirm that the dual-board setup is complete.

The use of the dual-detector is basically the same as the use of a single detector, the difference is that the dual panel has an additional dual panel switching function, making it more convenient to use.

After filling in the patient information, fill in the protocol to be selected, the protocol has been selected for each position to use the panel switch: the first step to select the position, the second step corresponds to the exposure of the flat detector switch, the figure shows the selected lying detector, if you want to select the standing flat panel detector, choose the icon of the body standing.

Finally click on the inspection, the specified body position will be exposed to the specified plate detector (Note: If you select a good position, also select the corresponding body position using which piece of detector exposure after clicking on the registration, from the work list into the patient exposure interface, the position corresponds to the exposure of the plate detector needs to be reset).

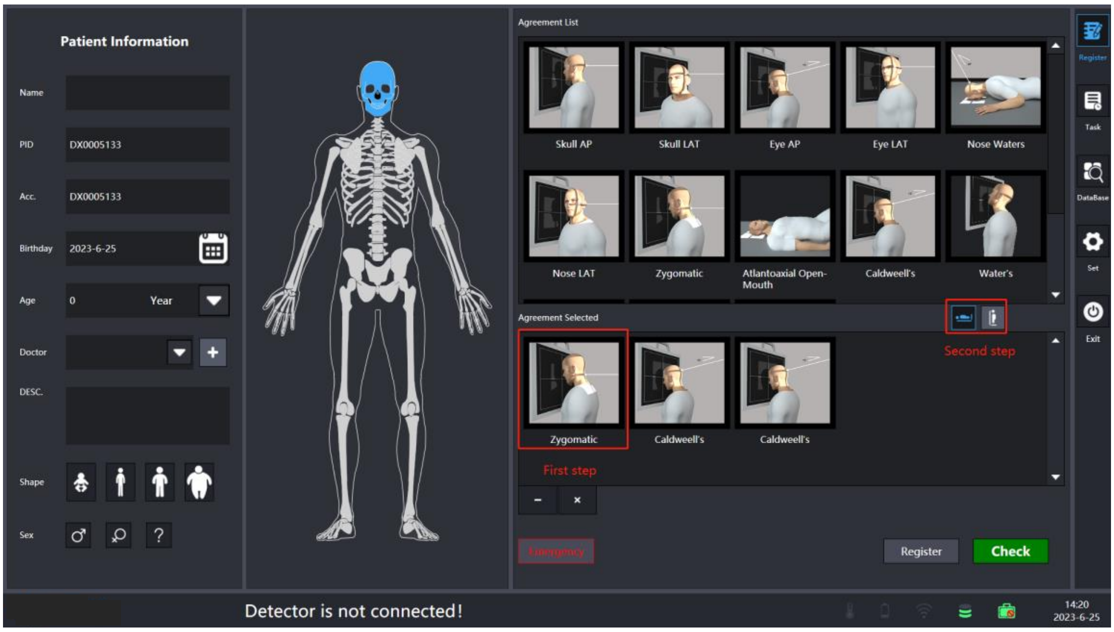


Figure 4. 14-2 Patient Information Input Interface

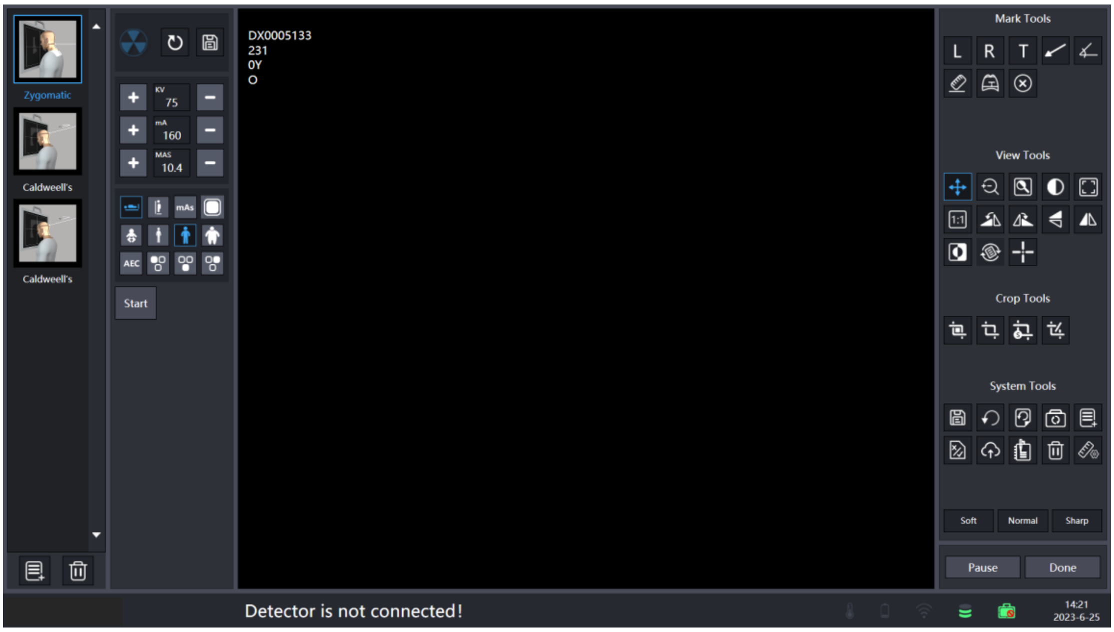
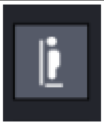


Figure 4. 14-3 The expose Interface

Switching the flat detector in the exposure interface can be done by simply clicking on the standing or lying icon.

Symbol	Description
	In the system setting, the "lie" corresponds to the detector.
	In the system setting, the "stand" corresponds to the detector.

4.15 Dual-Screen Usage

Enter the system settings interface, set the secondary screen to Enable, click to modify the confirmation, re-enter the exposure interface and other image processing interface, another monitor connected to the computer will open the secondary screen interface. Exit the image processing or exposure interface, the secondary screen will automatically close.

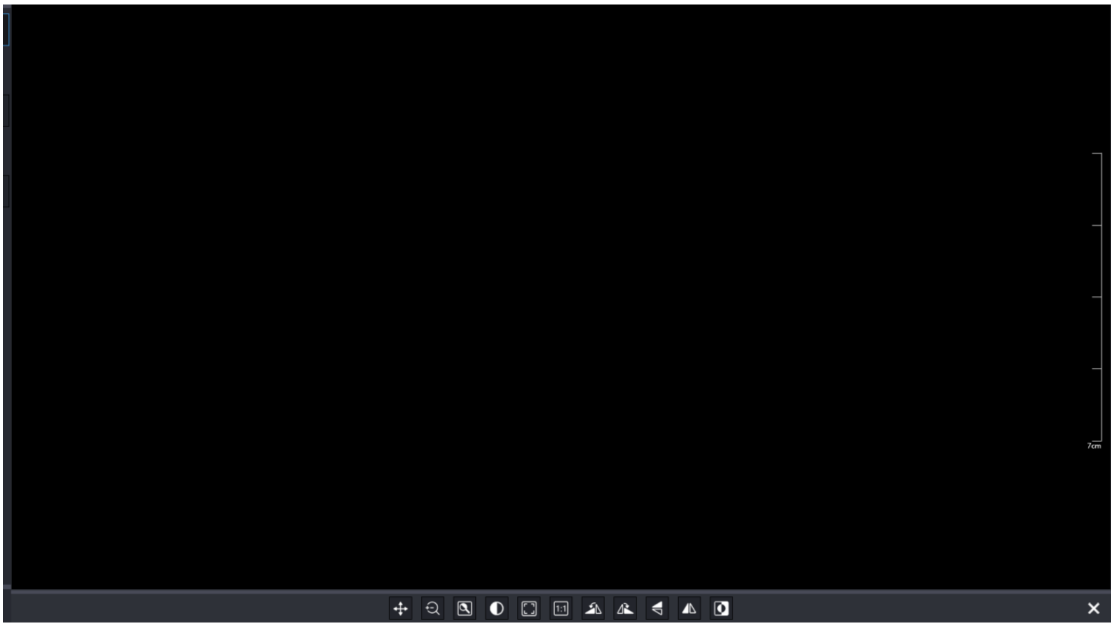
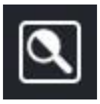
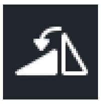
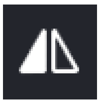
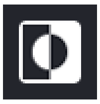


Figure 4. 15-1 Vice Screen Interface

Symbol	Function	Description
	Pan the image	Click this button, hold down the left mouse button in the image display area and drag, the image will be panned up, down, left and right as needed.
	Zoom image	Click this button and hold down the left click of the mouse in the image display area to zoom down on the image and zoom up on the image.
	Area of interest window level and window width	Click on the button, left mouse click on the image display area, from the upper left to the lower right to draw a rectangular area (rectangular area is not displayed), after releasing the mouse, the selected area of the window that is the area of interest suitable window width window level to display the image, and the overall window width window level is different, ROI window width window level is used to optimize the local image.

	Window Level/ Window Width	Click this button, left-click the image display area when dragging the cursor up, the window level decreases; drag the cursor down, the window level increases; move the cursor to the left, the window width becomes smaller; drag the cursor to the right, the window width increases.
	Adaptable button	Click this button then current image will zoom in or out automatically to fill all image area window.
	Real size	Click this button to display image in an actual size.
	Rotate 90 degrees counterclockwise	Click this button to rotate the image 90 degrees counterclockwise.
	Rotate 90 degrees clockwise	Click this button to rotate the image 90 degrees clockwise.
	Flip up and	Click this button to flip the image

	down	vertically and swap it up and down.
	Flip left and right	Click this button to flip the image horizontally and swap it left and right.
	Inverse	Inverse the image color.

4.16 Physical examination mode

4.16.1 Physical examination mode

Enter the setting, set the PE to "Enable", confirm the modification, and restart the software.

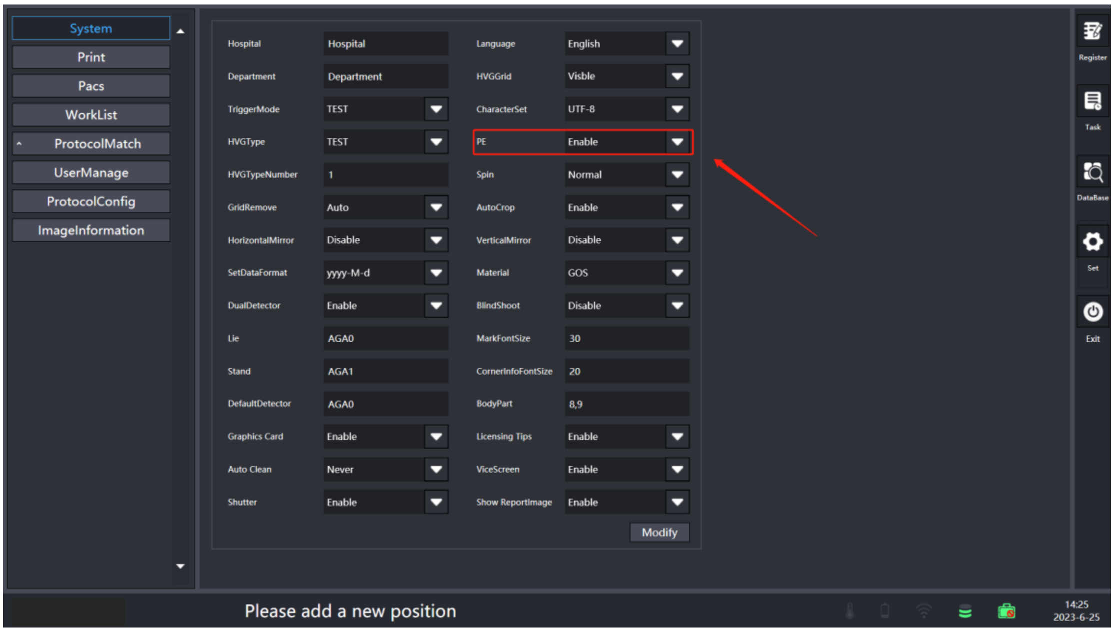


Figure 4. 16-1 Set the interface

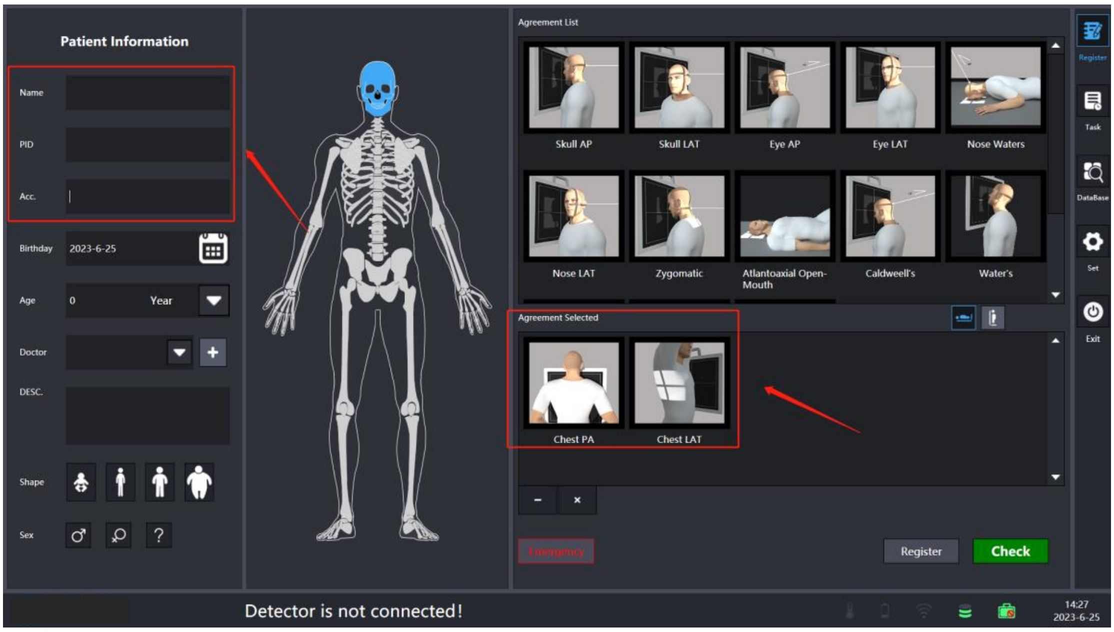


Figure 4. 16-2 Registration interface

In the PE mode, the registration interface number and the check number is empty for the first time. The cursor selects the check number by default, enters the check number to automatically match the number and name, and the default position protocol of the two Settings.

4.16.2 scanner gun

First, connect the scanning gun to the computer, and scan the QR code or bar code on the registration interface to identify the registration. (Scanning gun function is only effective in PE mode) As shown in the figure below:

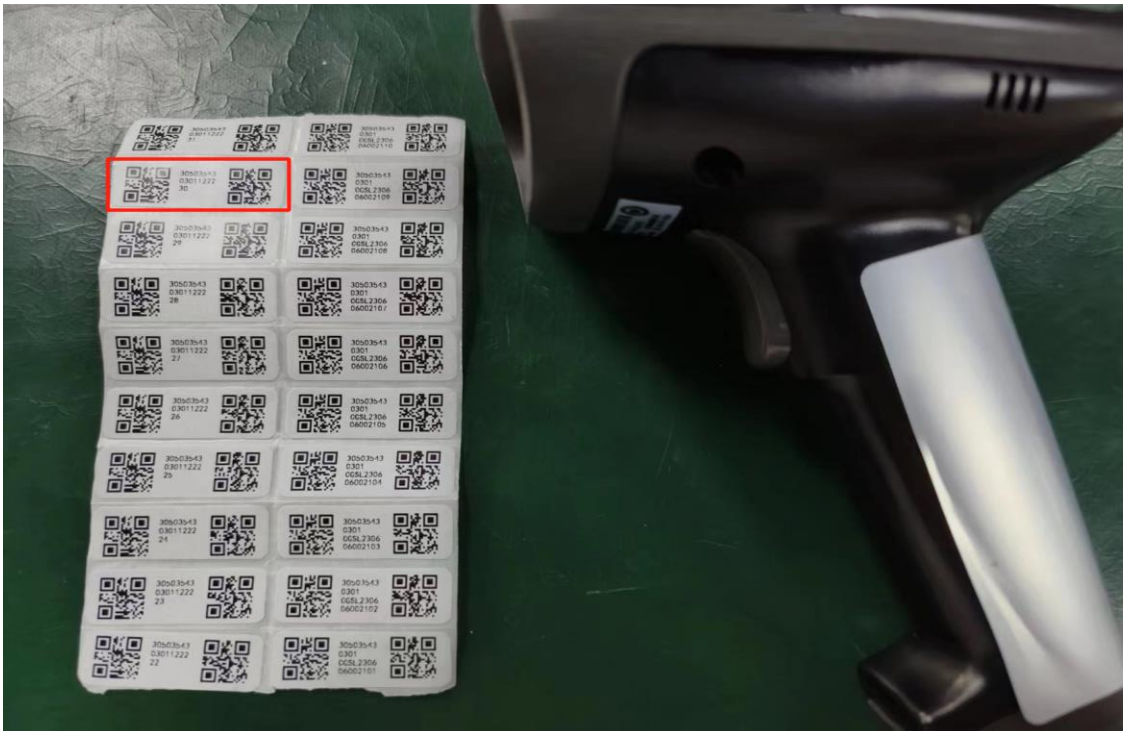


Figure 4. 16-3　A QR code and a scanning gun

Scan the information of the two code in the figure shown in the red box,automatically identify and fill in the text box of the check number.

Figure 4. 16-4 Registration interface

In the task interface, scanning gun registration qr code information, can directly enter the shooting interface for exposure, such as scanning unregistered qr code information prompt cartridge current scanning task does not exist, whether to register and open, according to the corner information into the case exposure interface and registration information.

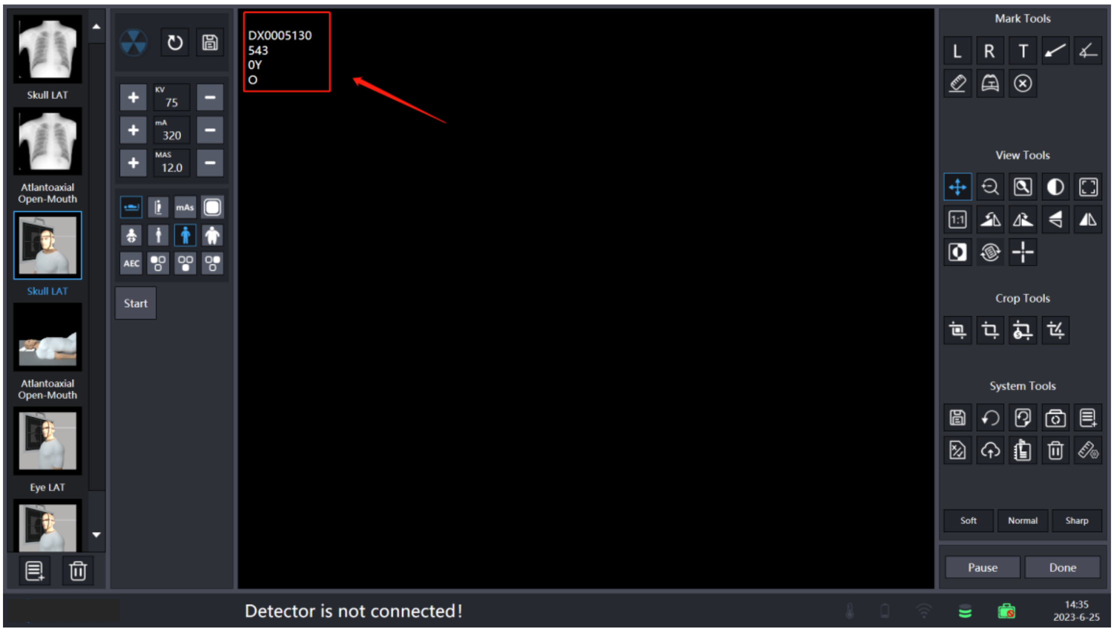


Figure 4. 16-4 Exposure interface

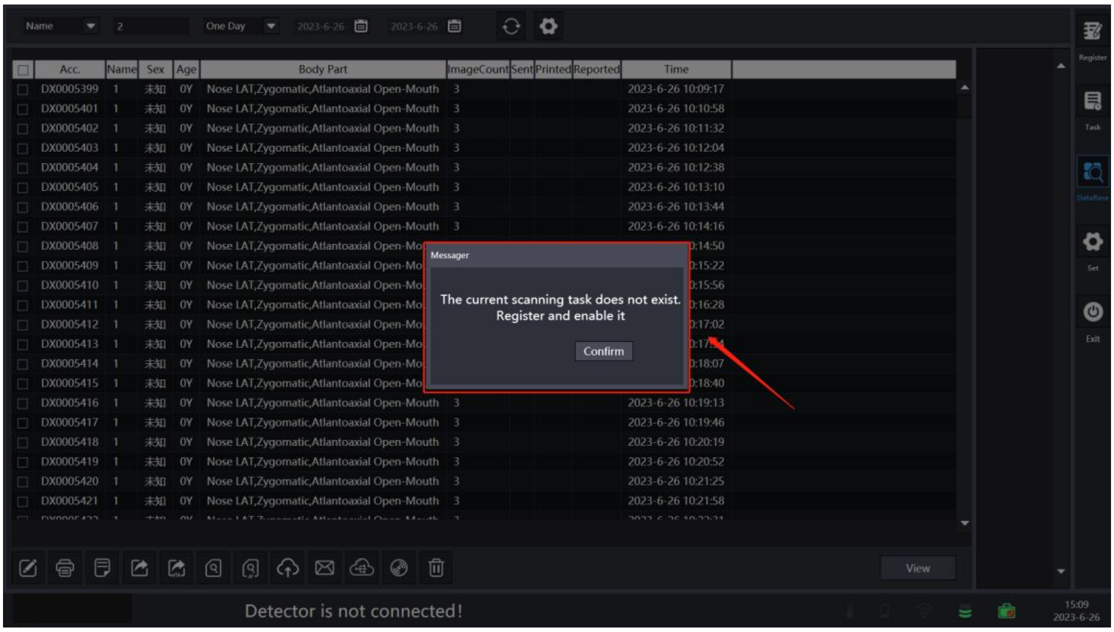


Figure 4. 16-5 Task interface

The exposure interface still has an unshot exposure picture. Click to complete or switch the case is indicating that the shot picture is unfinished, whether it is suspended:

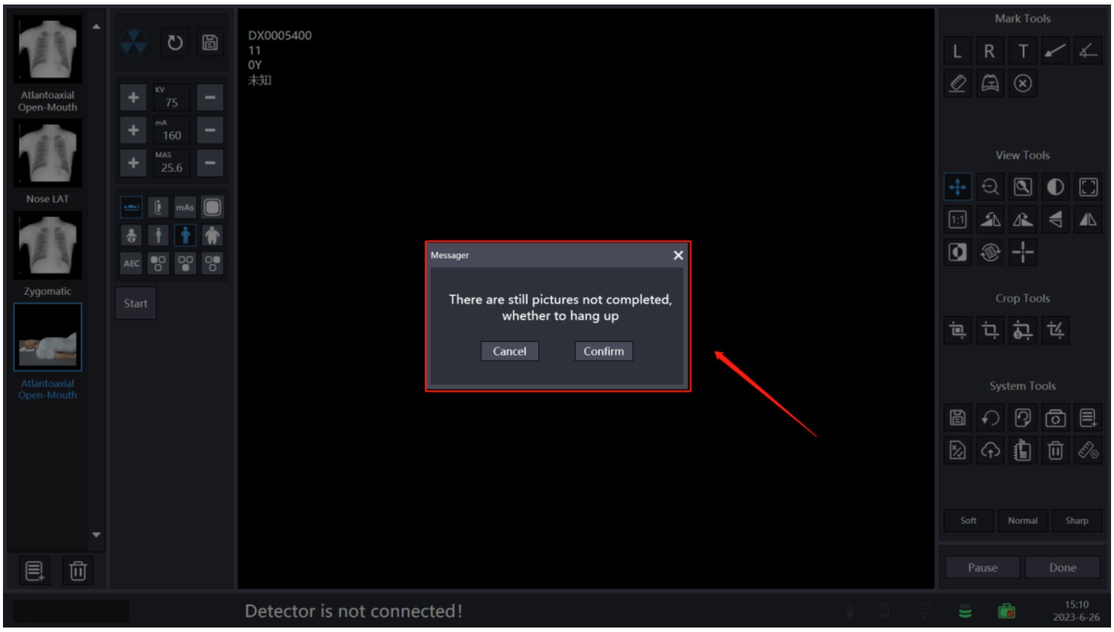


Figure 4. 16-6 Task interface

The completed case indicates that the pop-up exposure map has been taken and whether it needs to be opened. As shown in the figure below:

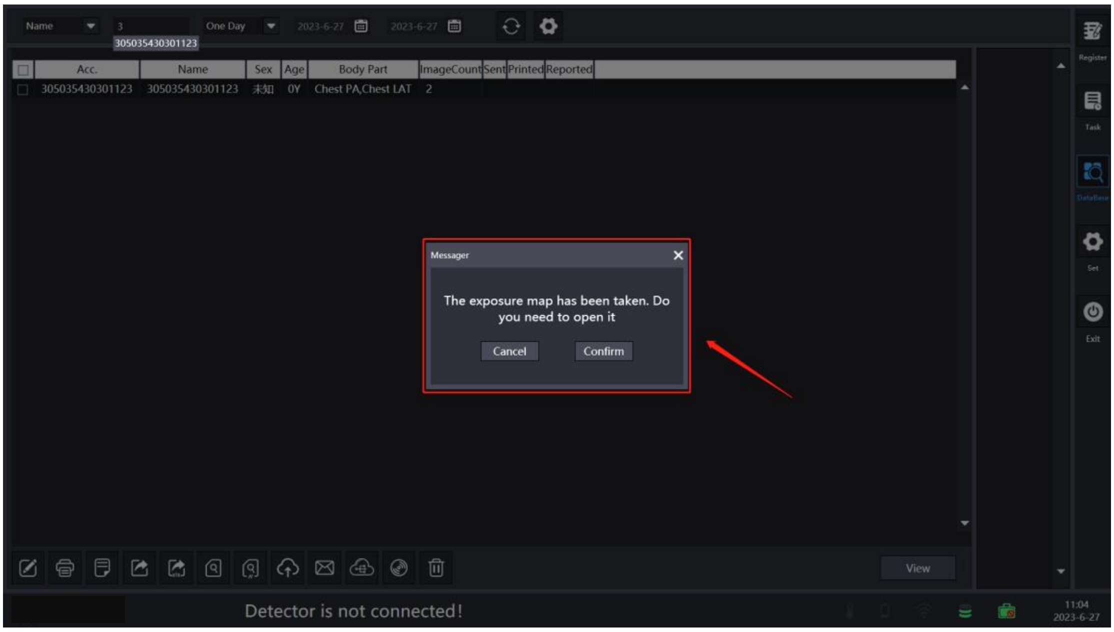


Figure 4. 16-7 Database interface

4.17 E-mall

Enter Settings, select the E-mail module, and select the message type used. As shown in the following figure:

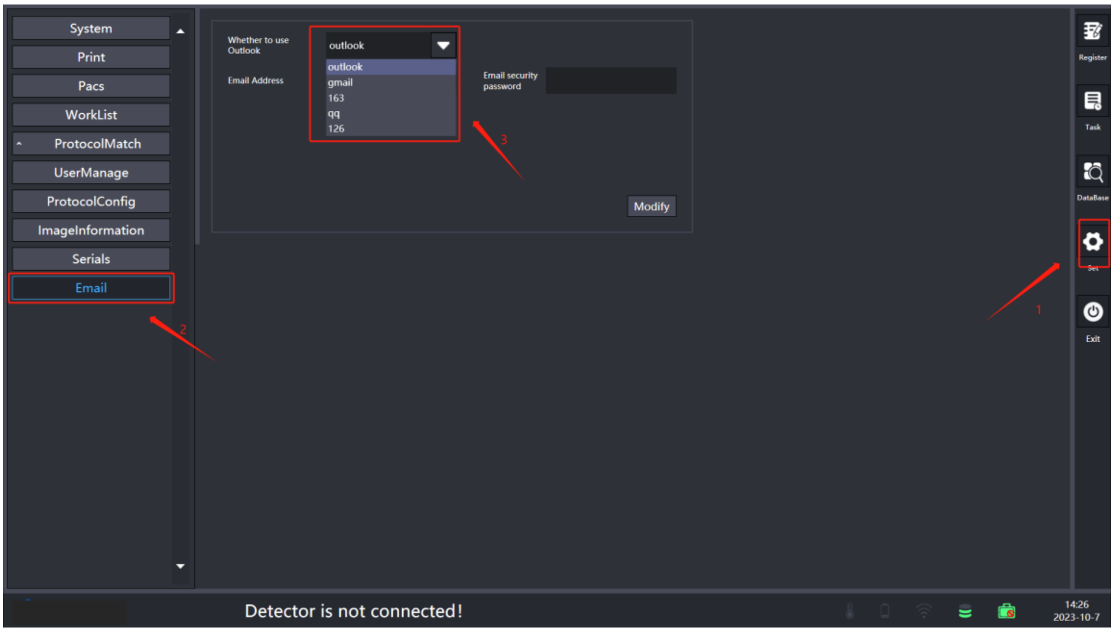


Figure 4. 17-1 Set the interface

Enter the email address of the sender and the email authorization password, click to modify. As shown in the following figure:

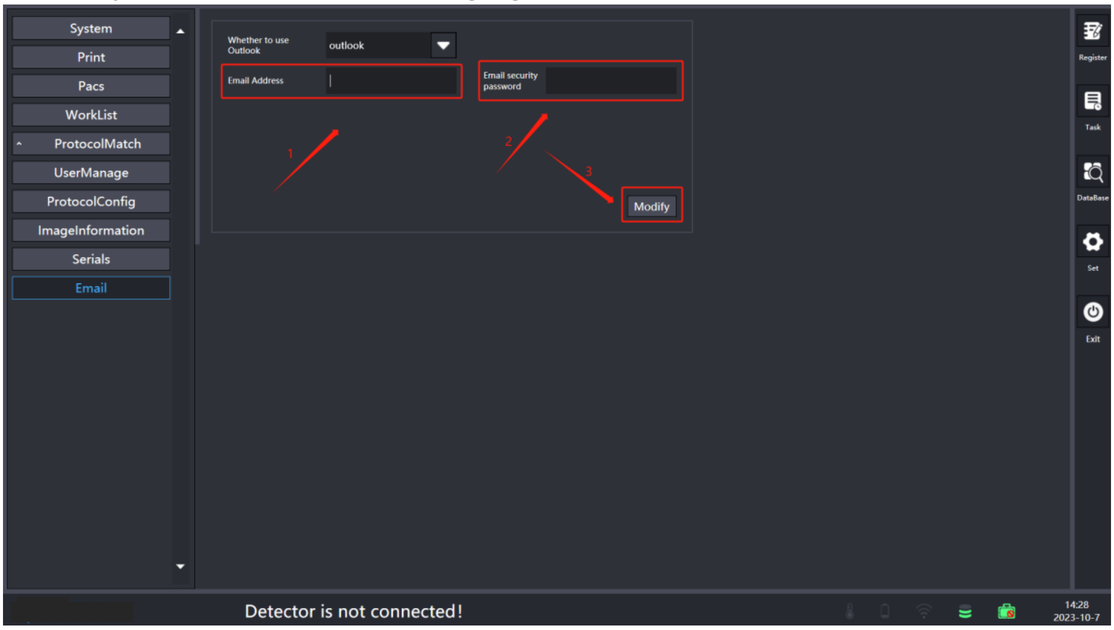


Figure 4. 17-2 Set the interface

To obtain the mailbox authorization code such as 163 mailbox, enter the 163 mailbox setting interface, POP 3 / SMTP / IMAP module, and use the authorization password management to obtain the authorization password.↳ As shown in the following figure:

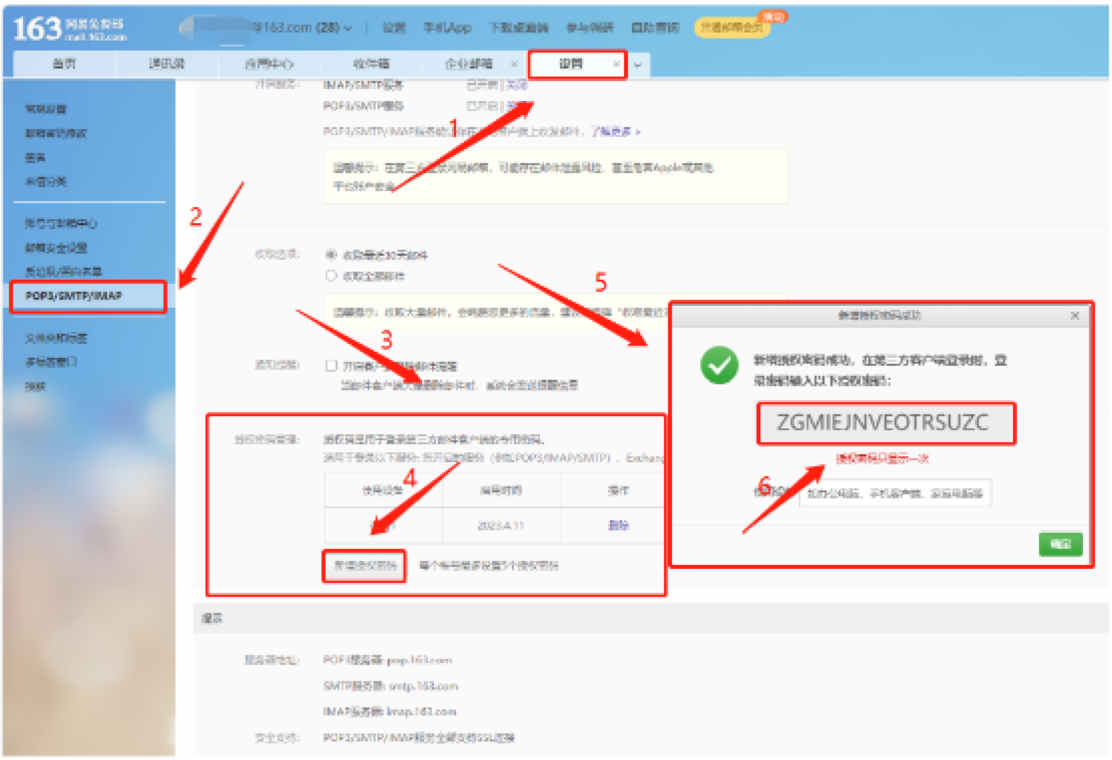


Figure 4. 17-3 163 Email setting interface

To obtain the email authorization code such as QQ mailbox, enter the QQ mailbox Settings interface, use the account number / POP 3 / IMTP / SMAP module, and use the authorization password management to obtain the authorization password. As shown in the following figure:

Figure 4. 17-4 QQ Email setting interface

To obtain the mailbox authorization code such as 126 mailbox, enter the 126 mailbox setting interface, POP 3 / SMTP / IMAP module, and use the authorization password management to obtain the authorization password. As shown in the following figure：

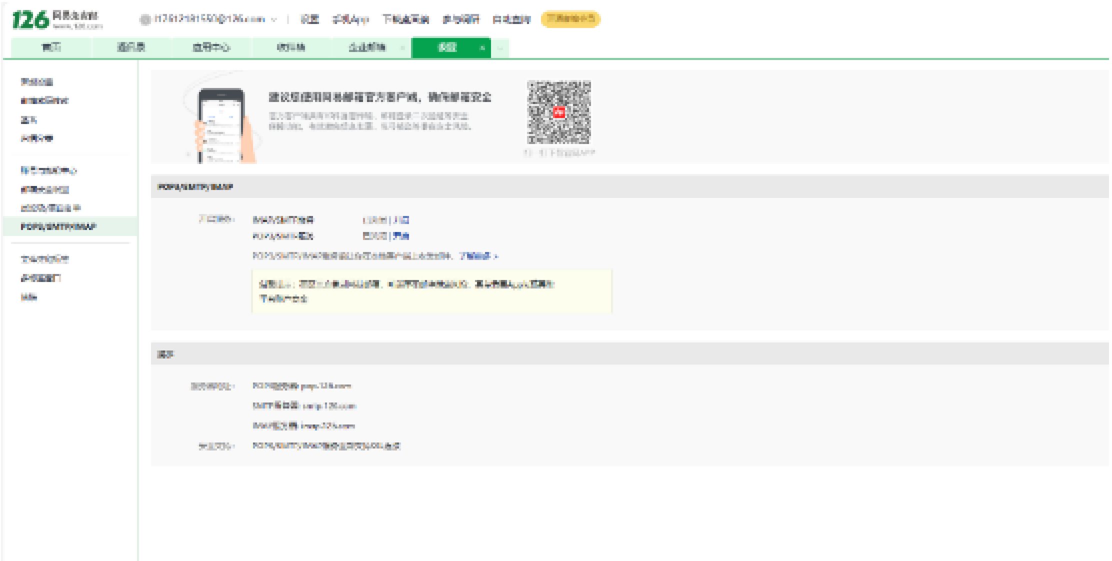


Figure 4. 17-5 126 Email setting interface

After the setting is complete, enter the database interface, select the case information to be sent to the email, enter the receiving user email information, click OK, and transmission is successful. As shown in the following figure:

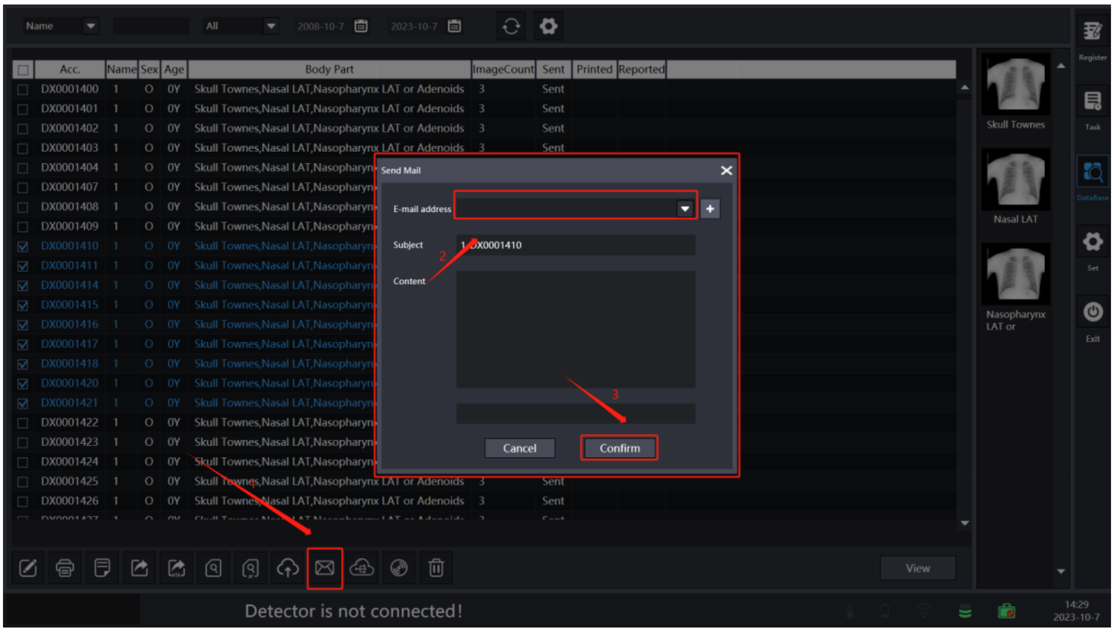


Figure 4. 17-6 Database interface

After the transmission is completed, enter the receiving mailbox to view the received case information. As shown in the following figure:

Figure 4. 17-7 QQ Email setting interface

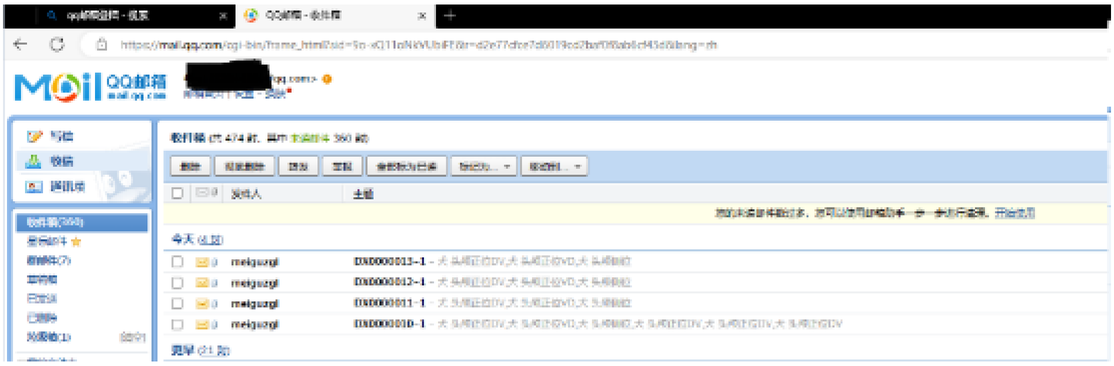


Figure 4. 17-8 QQ Email setting interface

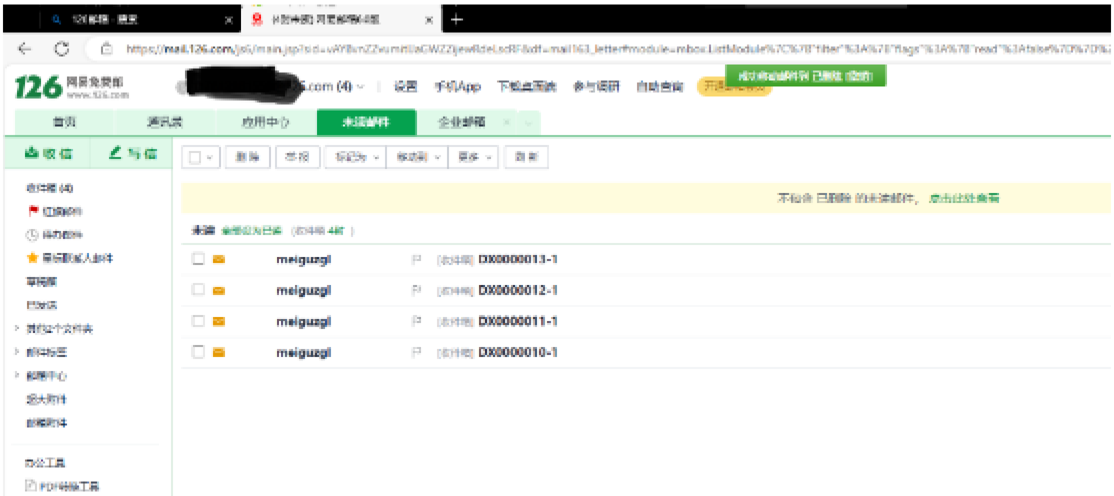


Figure 4. 17-9 126 Email setting interface

4.18 Automatic printing

4.18.1 Local printing of multiple cases

Enter the settings interface, select the Local Print option for automatic printing, and then click the modify button. As shown in the following figure:

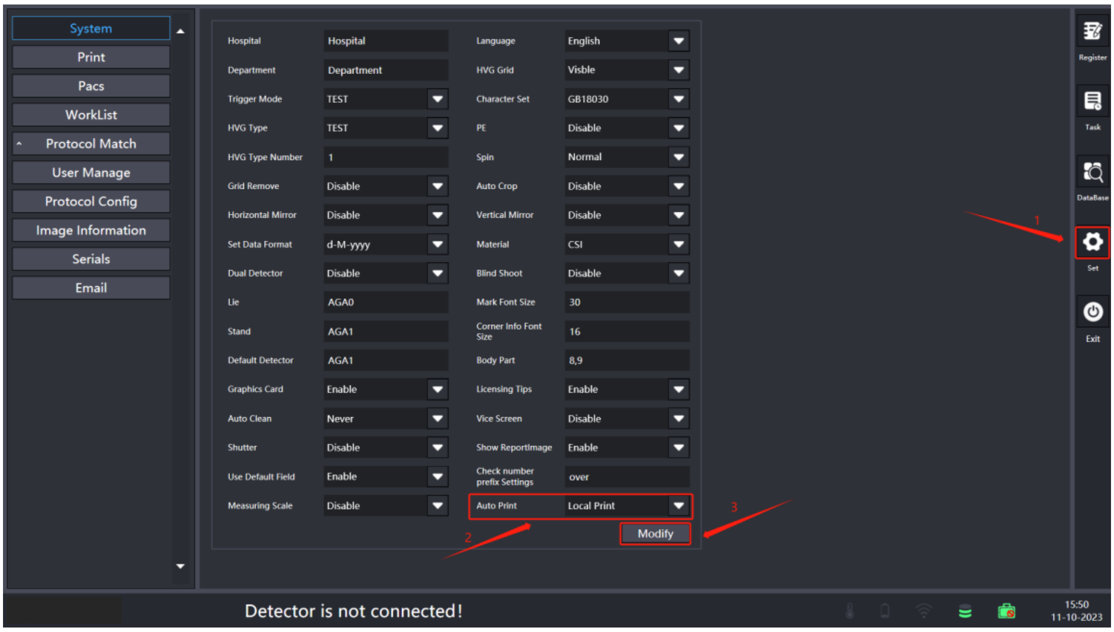


Figure 4. 18-1 Set the interface

Then enter the printing configuration interface, select the film size and printing direction in the local printing module, and click Modify Save to exit and restart the software. As shown in the following figure:

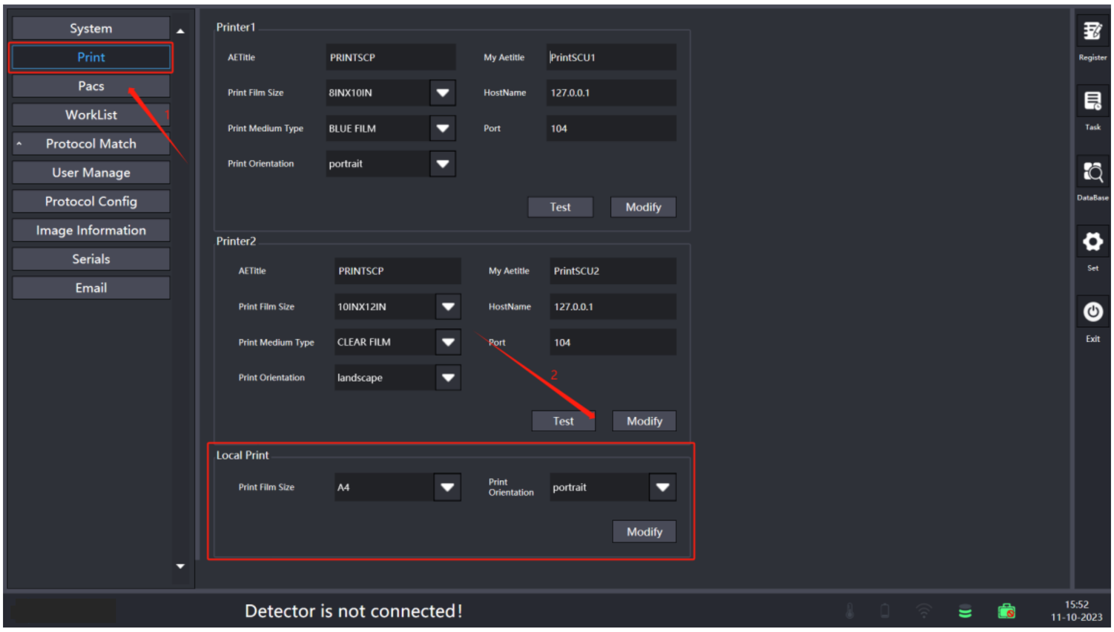


Figure 4. 18-2 Set the interface

Go to the database interface, drag to select or check the case that needs to be printed, and click the print button. As shown in the following figure:

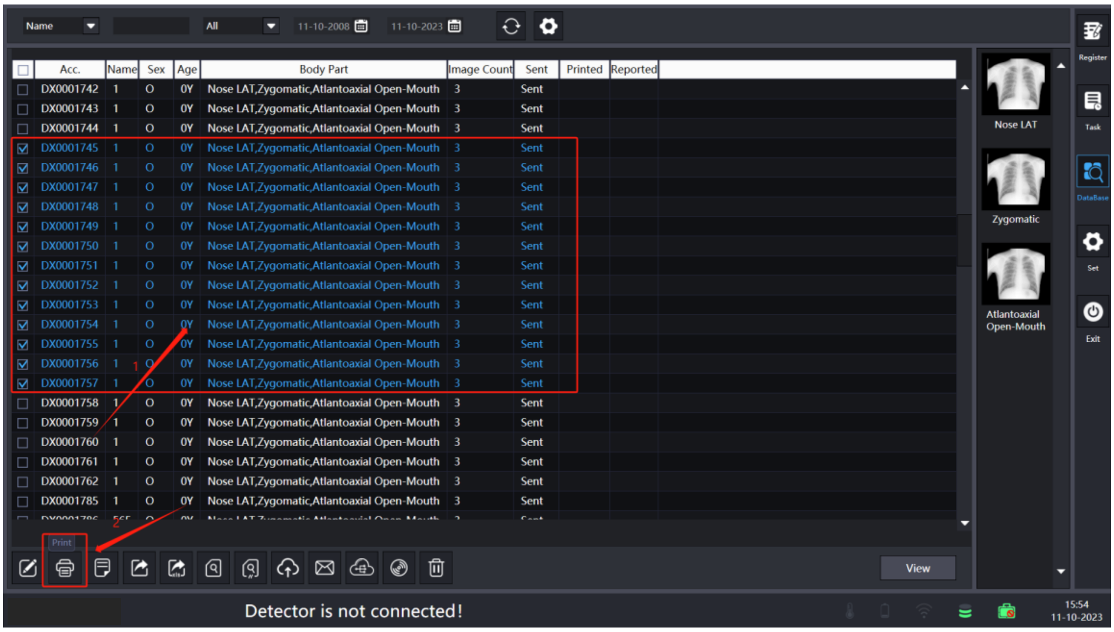


Figure 4. 18-3 Database interface

In the pop-up dialog box, select a local printer, print the direction, and then click Print. As shown in the following figure:

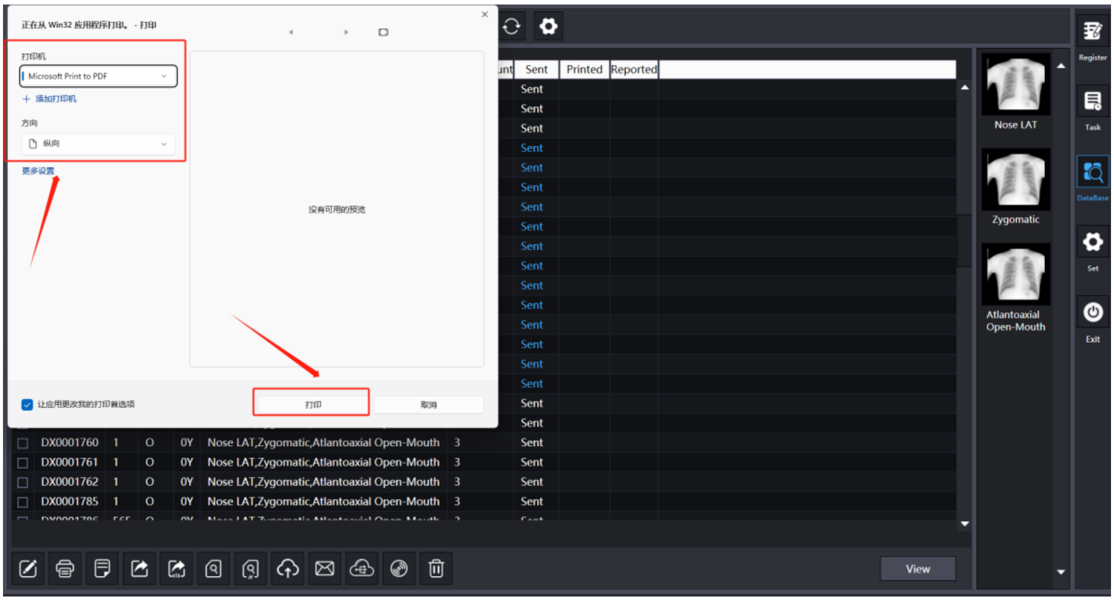


Figure 4. 18-4 Database interface

After printing is completed, a successful printing dialog box will pop up and click OK. The case printing is successful. As shown in the following figure:

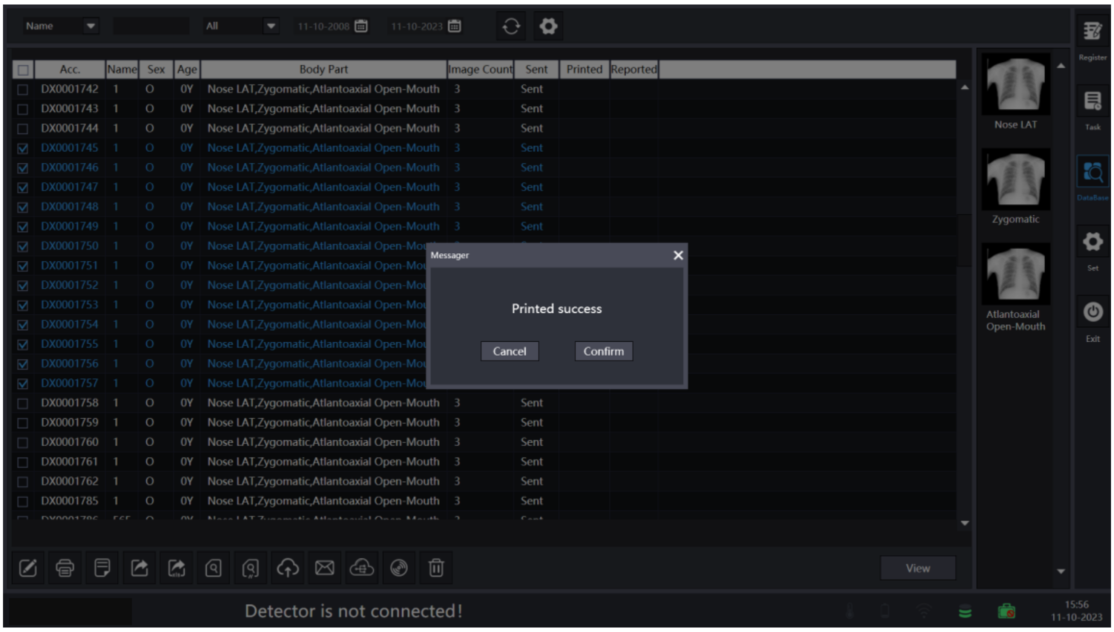


Figure 4. 18-5 Database interface